A Dated, Graded Ledger of the Testable Predictions of a Vacuum-Tied MOND Framework, with Its Confronted Record and Retracted Claims

Working note

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Abstract

We collect, in one dated ledger, every testable prediction we could find in the working record of a programme built on the law $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with $\kappa = \frac{1}{2}$ *fitted*, together with the numbers the rivals predict, the result that would count against each prediction, and what has already been confronted with data. Each of 98 future tests carries a grade: K1 (the number was fixed before the data exist, by frozen criteria or a sealed hash, with a date and a stated kill condition; 8 tests), K2 (sharp but capped by systematics, conditional, or nearly shared by the rivals; 11), K3 (open: no usable data, no date, or a forecast only; 65) and K4 (speculative or numerology-era predictions that could still be checked but have no derivation and are mostly retracted in the record; 14). We also list 51 confronted results, including the failures, 15 public deposits with their status, and the claims that must not be cited. The dated releases in the next months are the ALMA cubes of 2026-11-06 and 2026-11-28 and Gaia DR4 on 2026-12-02. The neutrino sector, which is entirely conditional and rests on a coincidence that the record itself calls generic, is set out in detail. Nothing here says the data favour the framework, κ is not derived, the cold energy's mass is still required, and the theory is not closed.

1 Scope and standing terms

The acceleration scale is $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted (the current estimate is 0.42 ± 0.10, CFG493). Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are never pooled and are written "can / alt". The cold energy's mass is still required and no dark-matter particle is proposed. Verdicts use raw data against framework-native predictions only; rows whose machinery is calibrated on ΛCDM carry no verdict weight. The flat law $a_0(z) = \text{const}$ is the framework's distinctive high-redshift law, $a_0 \propto H(z)$ is its rival, and ΛCDM has no a_0 (its emergent scale is the proxy).

Method. The ledger was compiled on 2026-10-11 from nine read-only sweeps of the programme's repository (the predictions ledgers, the Gaia DR4 pre-registration and its 21 amendments, lanes CFG1–CFG700, the atomos kill sheet, `real_research`, `hunt_2026`, the multi-model exploration folders, git history since April, and memory notes). Some lane folders were only keyword-screened, so the list is thorough but not guaranteed complete. Release dates for Euclid, Rubin, JUNO, LISA and 4MOST are taken from the record's own inventories and were not independently verified. Rows from AI exploration branches are tagged BRANCH and are non-authoritative; rows tied to the retired modified-inertia arm are tagged MI-ARM and are not live; rows from the old clock-host candidate are tagged CANDIDATE-CONDITIONAL.

Grades.

- **K1**, the Kepler grade: a number fixed before the data exist (frozen criteria or a sealed hash), a date, and a stated result that counts against it.
- **K2**: capped. The number is sharp, but the test cannot reach 3σ with today's systematics, the rivals predict nearly the same, or it is conditional on an assumption.
- **K3**: open. A real prediction with no usable data, no date, or only a forecast.
- **K4**, "Kepler slop": speculative or numerology-era predictions. Dated and testable, so they could still land, but no derivation backs them and the record has retracted or downgraded several. They are never to be cited as framework results.

2 Calendar of dated releases

2026-10-11 to 13	Framework-native cosmic shear, 1024^3 run (CFG596), verdict pending
2026-11-06	ALMA U4_27928, CO at $z = 1.42$ (sealed predictions, CFG571)
2026-11-28	ALMA GS4_24110, deep [CI] at $z = 2.00$ (sealed predictions; part already scored on other public data)
2026-12-02	Gaia DR4: wide binaries (pre-registration, 21 amendments) and Milky Way vertical force and curve (sealed, CFG553)
2026-12-17	ALMA Big Wheel at $0.24''$ (no sealed prediction)
2027–2028	Rubin SNe $w(z)$ gate, JUNO θ_{12}, DESI DR3
2030s	LISA, DUNE and Hyper-K, LEGEND-1000, CMB-S4

3 Future tests

Sorted by date within each group. Rows are generated from the same data file as the public chart, so the two cannot differ.

Dated releases

The first rows were sealed before their data became public.

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
2026-10-11 to 13 *1024^3 run in progress*	**Framework-native cosmic shear (CFG596)** Lean 1024^3 particle-mesh, L = 100 Mpc/h, snapshots z = 1, 0.75, 0.5, 0.26, 0	Own gravitating field against raw KiDS-1000 and DES Y3 $\xi\pm$. Diagnostic only if the framework's own boost gives $\chi^2 \geq 9$ on the surviving points (tooth T-OWN).	**Rivals.** Same-pipeline comparison model S0. Today's 512^3 runs lean against ($\Delta\chi^2$ +23.9 / +4.2 can, +30.0 / +5.6 alt on KiDS / DES). **Counts against.** A decisive version needs L $\geq$ 400 Mpc/h at N $\geq$ 1024. Verdict pending. *CFG596, CFG592*	K1
2026-11-06 *public release*	**ALMA U4_27928** CO at $0.07''$, z 1.419; required gas for the measured rotation	At V_c 225 km/s: log M_{gas} 9.84 / 9.46 (F-DESI) and 9.75 / 9.19 (F-flat) can / alt. At 250–300 km/s every model needs gas and the framework needs 0.2–0.9 dex more than $a_0 \propto H(z)$.	**Rivals.** At V_c 225 both rivals ($a_0 \propto H(z)$, ΛCDM+feedback scale) sit at the FLOOR: stars alone already exceed what they allow. Below 200 km/s every model is at the floor and the test says nothing. **Counts against.** Gas outside the required band at $V_c \geq 225$. One disc, so nothing is excluded. *CFG571, sealed hash in PREDICTIONS_HASH.txt*	K1
2026-11-28 *proprietary until*	**ALMA GS4_24110** Deep [CI](2–1) at $0.09''$ and dust, z 1.997 (2025.1.01377.L)	Required log M_{gas} (can, M* 10.89, $R_g = R_d$ / $1.5R_d$): F-flat 9.5 / 9.7; F-DESI 9.8 / 10.1. Alt footing F-DESI 9.1 / 9.4. The stellar-mass branch moves it by 0.6 dex.	**Rivals.** $a_0 \propto H(z)$ and ΛCDM+feedback at the FLOOR in every branch. **Counts against.** Gas near 10^{10} M$\odot$ inside R_e counts against both rivals; negligible gas troubles the dark-energy-tracking branch. **Note.** Already partly scored on other public data (CFG572/573): CO gives log M_{gas} 10.95, dust 10.5–10.7, [CI] below 10.3. The framework fits only with the dust gas, tracers disagree by about 0.7 dex, no model is excluded, and CFG573 was not diagnostic. *CFG570, CFG571, CFG572, CFG573*	K1
2026-12-02 *Gaia DR4*	**Wide binaries: candidate B (Arm C)** Velocity boost $\hat{\gamma}$ at 2–30 kAU, about 30,000 pairs; pipeline and 21 amendments frozen	Hierarchical ownership gives $\gamma = 1.000$ exactly, no a_0 dependence. 0.944–1.056 is consistent; 1.056–1.084 disfavoured at 2–3σ_{tot}.	**Rivals.** Newton and ΛCDM: 1.000, the same. A Newtonian result kills the strict-MOND arm and cannot confirm B. **Counts against.** $\hat{\gamma} \geq 1.084$ with the stability checks passing kills B (and Newton and ΛCDM with it). Report raw $\hat{\gamma}$ with σ_{fit} and distances to every anchor, never one verdict word. *PREREGISTRATION_DR4.md Amdt 13, 21; CFG63*	K1
2026-12-02 *Gaia DR4*	**Wide binaries: strict MOND + EFE (Arm A)** Full AQUAL external-field solve (Amdt 10)	$\hat{\gamma}$ = 1.1614–1.1814 can; 1.1917–1.2267 alt. Newton sits 5.8σ_{tot} below the canonical floor at N = 30,000.	**Rivals.** B, Newton, ΛCDM: 1.000. **Counts against.** $\hat{\gamma}$ below about 1.056 falsifies it. Above 1.23 is unscoreable (contamination guard); 43–49% of true alt-top outcomes land there. *Amdt 10, 11; CFG63*	K1

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
2026-12-02 *Gaia DR4*	**Wide binaries: derivation-chain law and merge band** Amdt 14 (filter length ξ); CFG200 forecast (provisional)	Chain ceiling 1.0725 / 1.0900, falling to 1.000 by $\xi \geq 0.3$ pc. Door-11 merge band 1.089–1.102 (g_{ext} 1.778e-10) or 1.063–1.077 (2.146e-10).	**Rivals.** Newton / B / ΛCDM 1.000. Chain vs Newton: 3.6σ can / 4.5σ alt at the ceiling, needing N = 58,850 / 21,660 for 3σ. PAPER38 v1.3 gives the framework's own P2 kernel as 1.063–1.102 can (1.079–1.127 alt), 2.3–3.7σ from ownership at N = 30,000. **Counts against.** Chain killed at $\geq$ 1.157 / 1.174. Merge vs ownership 3.2–3.7σ at N = 30,000 (alt 4.0–4.6σ). *Amdt 14; CFG200*	K2
2026-12-02 *Gaia DR4*	**Wide binaries: Arm B (covariant) and anisotropy** Amdt 12, 20	Arm B: 1.0000 ± 0.0025 ($\xi \geq 4$ pc), flagged not a testable arm (cap 2.25σ can / 1.50σ alt). Arm A anisotropy: perpendicular-dominant, +0.0013 to +0.0046.	**Rivals.** Outer pairs (beyond about 1.96 r_M) parallel-dominant is expected and not a kill. **Counts against.** Arm B killed $\geq$ 1.084. Opposite-sense anisotropy at $\geq 3\sigma$ falsifies the derived EFE; a null is not a kill. Quoted only if neither fit sits on a grid edge. *Amdt 12, 20*	K2
2026-12-02 *Gaia DR4*	**Milky Way vertical force and local dark density** Five sealed observables (CFG553); round cold energy against the QUMOND phantom disc	$K_z(R_0, 1.1)/2\pi G$ 74.1 (68.3–79.9) / 75.6 M⊙/pc^2; local dark density 0.323 / 0.360 GeV/cm^3; Oort limit 0.116 / 0.117 M⊙/pc^3; $\sigma_z(R_0)$ 16.8 / 16.9 km/s.	**Rivals.** Phantom disc: K_z 90.1 / 94.8, dark density 2.01 / 2.34, Oort limit 0.160 / 0.169, σ_z 19.3 / 19.8. A fitted NFW gives K_z 76.3, so the round rule cannot be separated from NFW here. **Counts against.** K_z near 90 favours the disc. Needs 3.5–5.8 M⊙/pc^2 on one K_z value; the 36-cell K_z(R, z) grid separates at 40–80% per-point error. Round beats the disc in 32/32 cells today. **Note.** Control K6 failed as frozen (a mis-specified expectation); the spherical-equivalent column was added before the seal and disclosed. *CFG553, hash f8bea44f... in PREDICTIONS_HASH.txt; CFG514/516*	K1
2026-12-02 *Gaia DR4*	**Milky Way outer curve** Distance scale at 15–27.5 kpc decides	dlnV/dlnR 15–22 kpc -0.162 / -0.149; V(20 kpc) 199.5 / 206.6 km/s; V_c(60 kpc) 178.1 / 186.0 km/s; mean V_c 30–60 kpc 182.6 / 190.3.	**Rivals.** NFW fit: slope -0.171 to -0.174, V_c(60) 166.5. Round and disc share the in-plane speed, so this cannot separate them. **Counts against.** A slope of -0.25 or steeper at $\sigma \leq 0.04$ excludes both the framework and NFW. V_c(60) near 166 favours NFW. Today's shape test already reads Z -3.6 / -3.9 (see confronted). *CFG553 P2, P5; CFG532b*	K1
2026-12-02 *Gaia DR4*	**Stream flattening and $K_z(z)$ shape** GD-1 / Pal 5 at 10–20 kpc	Round-rule shapes from CFG514, with ASSUMED DR4 errors.	**Rivals.** Phantom disc. **Counts against.** $\sigma(q_\Phi)$ 0.05 gives about 2σ; $K_z(z)$ shape 5σ at 3% bins. *CFG514*	K3
2026-12-17 *public release*	**ALMA Big Wheel** CO(4–3) at 0.24″, one z 3.245 disc, 30 GB (2025.1.00107.S)	Nothing is sealed. A single object near y ≈ 2–4 at 10 kpc.	**Rivals.** Not forecast. **Counts against.** No frozen rule yet; a single object cannot be conditioned (CFG240), so it would be a labelled envelope. *CFG268*	K3

$a_0(z)$ and dark energy

Flat a_0 is the framework's distinctive law; $a_0 \propto H(z)$ is the rival; ΛCDM has no a_0 and its emergent scale is the proxy.

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
no date *needs new observations*	**a_0 at z ≈ 2.5** Lensed rotators, JWST IFU plus ALMA Band 3 CO(3–2); pre-registered in PAPER7 (DOI 10.5281/zenodo.22563139)	Flat: BTFR zero-point shift 0.00 dex.	**Rivals.** ΛCDM-native +0.334 dex (feedback-ΛCDM g† ×2.8 at z = 2); $a_0 \propto H(z)$ +0.58 dex; DE-tracking 0.09 dex below flat (e.g. $a_0(2.5)/a_0(0)$ = 0.755 for (w_0, w_a) = (-0.827, -0.75): -0.031 dex, 7.07e-11 can / 8.52e-11 alt). A robust -0.03 dex would confirm that branch and kill the flat law. **Counts against.** Offset $\geq$ 0.5 dex kills flat a_0; under 0.07 dex rules out H(z) tracking at 2σ (N $\geq$ 10). Needs 3 rotators at ±0.10 dex or 4 at ±0.20, with y from 3 down to 0.3 at 5% velocity. Cap from the mass-scale floor: 2.3σ against H(z), 1.3σ against ΛCDM. No object passes the gate today. *CFG52/54/63/385; CFG268; PAPER7*	K2

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
2027+ *Rubin / LSST-DESC SNe*	**Dark-energy w(z) gate** Frozen 2026-07-21 (DOI 10.5281/zenodo.21478568)	$a_0(z)/a_0(0) = \sqrt{(\rho_{DE}(z)/\rho_{DE}(0))}$, κ cancels. R = $a_0(3)/a_0(0)$ about 0.60–0.75; DESI DR2 baseline 0.775 [0.683, 0.878], a 2.0σ decline.	**Rivals.** Constant a_0 (R = 1); $a_0 \propto H(z)$. **Counts against.** OPEN if $\geq 3\sigma$ with R in [0.55, 0.85]. DISSOLVED to constant-a_0 if flat within 2σ (not a kill). STRAINED if R > 1 at $\geq 3\sigma$. Rubin reaches 3.3σ on forecast. Today UNDECIDED. *rubin_prereg; ledger A-3*	K1
DESI DR3 (2026–27) *date not on record*	**Dark-energy evolution, framework prediction** CFG549, sealed (hash f110566e… in PREDICTIONS_HASH.txt)	The framework does not predict $\rho_{DE}(z)$. It gives Λ plus a settling trickle $\Delta\rho/\rho \approx$ 7.1e-7 / 2.0e-7 (can A / B), 1 + w(0) = -5.3e-8. Any real evolution is free.	**Rivals.** DESI's evolving fits; the ΛCDM tension is shared. **Counts against.** The galaxy channel needs N ≈ 1730 / 640 / 97 / 56 discs per bin at 0.1 dex for z = 0.5 / 1 / 2 / 2.5 to match DESI+DESY5. Verdict TENSION, same as ΛCDM, not excluded. *CFG549*	K2
no date *untested on data*	**Elastic-vacuum fork: a_0 follows -p_{DE}** Rigidity left free; one-modulus form excluded ($\Delta\chi^2$ +416)	$a_0(1)/a_0(0)$ = 1.16 / 1.39 / 1.26 / 1.94 and $a_0(2.5)/a_0(0)$ = 1.02 / 1.15 / 1.07 / 1.46 for DESI+Pantheon+ / Union3 / DESY5 / DESI+CMB.	**Rivals.** Two-component law H2: 0.83 / 0.78 / 0.80 / 0.73 at z = 2.5. Gap 0.09–0.17 dex. **Counts against.** A result at the H2 values rather than the elastic ones. Needs a_0 at z ≈ 1 and 2.5 to 0.1 dex. *CFG511*	K3
no date *older, CPL-era wording*	**DESI DR3 decision matrix** Pre-registered fleet note; re-check against the flat law before citing	Bump of +6.0% for (w_0, w_a) = (-0.727, -1.05), $a_0(z=3)/a_0$ = 0.648; +3.6% and 0.775 for (-0.838, -0.62).	**Rivals.** w = -1 dissolves it. **Counts against.** Can kill, never confirm. Direct detection of the bump is below the current floor (> 2030). *PREREGISTERED_EMPIRICAL_FLEET_2026-07.md*	K3
no date *needs new simulations*	**Emergent a_0 against Λ** Varied-Λ hydro runs, none exist	d ln a_0 / d ln Λ = 0.5.	**Rivals.** ΛCDM estimate -0.014 to +0.161. Time version: feedback-ΛCDM g† rises ×1.2 / 2.8 / 4.9 at z = 1 / 2 / 2.5 against flat (CFG566 corrected it to ×1.3–1.5, power $\leq$ 0.06). **Counts against.** Spec: runs at Λ_0 and $10\Lambda_0$, RAR to g ≈ 1e-12 m/s^2, error under 0.08 dex per run. *CFG545, CFG565, CFG566*	K3
no date *ALMA 2026.1.00363.S, not yet observed*	**KURVS CO gas survey** Seven discs with $v/\sigma_0 \geq 1.5$ at about 1.6″	Required baryons lower than the H(z) rival by 0.19–0.31 dex (Burkert and no-pressure branches, never pooled).	**Rivals.** $a_0 \propto H(z)$ at the FLOOR in 6 of 7 discs on the no-pressure branch. F-DESI vs F-flat 0.02 dex or less. **Counts against.** Forecast 1.0–1.6σ with a 0.15 dex floor: it would lean, not decide. Pressure support is the largest open input. *CFG571*	K2
no date *MIGHTEE-HI / LADUMA*	**Local a_0 slope with redshift** Anchored slope $a_0(z)/a_0(0)$ = 1 + 5.59z (can)	Flat: ×1.00. Anchored: ×1.11 / 1.31 / 1.50 at z = 0.02 / 0.055 / 0.09.	**Rivals.** $a_0 \propto H(z)$: ×1.010 / 1.027 / 1.045. **Counts against.** Not possible in 12 of 13 cells; the rival differs from flat by at most 4.5% at z $\leq$ 0.093. About 290 galaxies would be needed on statistics alone. *CFG258, CFG300*	K3
no date *needs a rotator at z $\geq$ 10*	**BTFR zero-point deficit at high z** Resolved rotators at z $\geq$ 10 (ALMA [CII], JWST)	Deficit at fixed baryonic mass: 0.009% z=5; 0.34% z=10; 2.6% z=15; 7.8% z=20; 19% z=30.	**Rivals.** Flat a_0 (zero); the MOND regime shrinks as Σ† falls. **Counts against.** Offset below z = 5 above 0.1% in v. Power about zero today. *SWEEP3 inventory (check before citing)*	K3
no date *JWST / ALMA direct gas*	**High-z BTFR footing fork** Constant a_0 (can) against a_0 rising as E(z) (alt)	Jeanneau-subsample offset: can 0.000, alt -0.243.	**Rivals.** Combined today 0.99σ (alt side) / 0.95σ (can side): a wash. **Counts against.** Needs a coherent gas term $\leq$ 0.10 dex. *jeanneau_refit; highz_tfr_fork*	K3

Galaxies, lensing and clusters

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
Euclid DR1 *date not confirmed; one inventory says Oct 2026*	**Cluster-lensing kink: exact null** Filed 2026-07-16; the detection target was retired (below the usable radius and the 0.06 dex floor)	Elastic-medium branch: kink at g_{bar} = 2.7092e-10 / 3.2707e-10 m/s^2 (y_c = 2.894); slope break -0.138 / -0.275.	**Rivals.** Pure MI and ΛCDM: no kink. **Counts against.** Rule D: $\geq 5\sigma$ break with slope change in -0.10 to -0.30. Rule N: p $\leq$ 1.7–2.5% per bin excludes the n = 1 throttle at 3σ. A null does not separate MOND from ΛCDM. *cluster_kink_spec; ledger A-6*	K2

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
Euclid / LSST *mid-2020s to 2027*	**Isolated-lens RAR to 1 Mpc** No accreted dark charge	Lensing stays on $\sqrt{(g_{bar}\ a_0)}$; χ^2/dof 2.03 / 0.94 on KiDS today.	**Rivals.** An accreting cold component multiplies the 100 kpc–1 Mpc signal by 3–20. **Counts against.** A ×2–3 excess at 100–300 kpc. Today's amplitude budget (±0.3 dex) still allows ×2. *ledger A-7*	K2
no date *needs DES Y3 ΔΣ (download declined)*	**Early-type shortfall is physical** KiDS early types about 30–40% short at 50–100 kpc	The shortfall persists in an independent survey.	**Rivals.** A KiDS or SLUGGS systematic. **Counts against.** Vanishing in DES or DESI-lens data. Replication with DESI DR1 lenses is the crispiest scoped step (CFG314). *KILL_SHEET G9; CFG314*	K3
no date *needs multi-epoch dispersions*	**Ultra-faint dwarfs: bare law against B's cold-mass rule** Only 2 multi-epoch systems today	Separation 0.38 dex between the isolated law and the rule.	**Rivals.** ΛCDM not separable here. **Counts against.** With 8 systems: median offset above the law > 0.35 kills the isolated law; < 0.03 kills the rule. Cap 2.5σ (2.1σ with the binary shift); 3σ never. *CFG42/46/63*	K2
no date *X-ray group masses*	**Groups at R500** Hydrostatic allowance 20% (0.079 dex) is the floor	B against the closure rule: Δ = 0.149 dex today.	**Rivals.** B vs bare law: Δ = 0.012 dex, not separable. **Counts against.** A shortfall above 0.17 dex (2σ) kills B's cold-share rule. Cap 1.8σ (2.6σ at R2500). *CFG63/81/85*	K2
no date *needs a bigger sample*	**Directional external-field effect** Aligned outer-curve asymmetry toward the attractor, WALLABY / MaNGA	Rule R7 (no external field): amplitude 0.	**Rivals.** AQUAL-type EFE +1–4%; elastic branch 0.5–0.77 of AQUAL. On record: n = 16 gave +2.95; WALLABY n = 25 gave -1.70 ± 2.12 (an amplitude). **Counts against.** Detection at the AQUAL amplitude needs N ≥ 560 and counts against R7; a deep null below 0.7% needs about 1,160. AQUAL vs the elastic branch needs about 6,000. *ledger A-4; aligned_firing*	K3
no date *BIG-SPARC not public*	**Environment independence of a_0** Per-galaxy a_0 tracks ρ_Λ, not local density	No correlation; the decisive null is 13–34σ on 175 SPARC galaxies already.	**Rivals.** Density-dependent a_0; cosmic-web slope: alt -0.177, Verlinde-style +0.5 (canonical 0, needs N ≈ 490, have 52–55). **Counts against.** a_0 correlating with local density. *ledger A-8; cosmicweb_a0*	K3
no date *BIG-SPARC, TRGB distances*	**Gas-dominated a_0 line** $E = g_{obs}^2 - g_{bar}^2 = a_0\ g_{bar}$	Gas slope 1.181e-10 ± 16% (GLS), 0.973e-10 (median); TRGB subset 1.333e-10.	**Rivals.** Footing not decided: both priors tighten but stay non-diagnostic. **Counts against.** A gas slope of 3e-10 breaks it outright. *a0_line, a0_line_trgb*	K3
no date *long-period comets*	**Oort-cloud anisotropy** Keyed to the galactic field direction (DOI 21966646)	Spike modulation 1.069 (7%); injection modulation 1.169 (17%).	**Rivals.** Isotropic injection. **Counts against.** Isotropy at the 5% level, or this and the wide-binary result disagreeing on ν_0. Prior art owed (Pauco & Klacka 2016). *ledger A-5*	K3
no date *SPARC + TRGB distances*	**$\kappa = \frac{1}{2}$ against $1/(2\pi)$** κ is fitted	Fit leans $\frac{1}{2}$ at about 2.2σ (one shape) / 1.55σ (shape-free); κ = 0.42 ± 0.10 (CFG493, can).	**Rivals.** $1/(2\pi)$; Gaia DR4 is not a κ-meter ($d\ln\gamma/d\ln\kappa$ = 0.18). **Counts against.** $\frac{1}{2}$ excluded at 3σ by the distance-free estimator (floor 3.9%). No shape reaches 3σ today. *ledger A-10; CFG493*	K3
no date *large clean RAR sample*	**Kernel fingerprint** The test is a_0-independent; this is Milgrom's kernel shape, not unique to the framework	At D = $\sqrt{2}$ the slope is 0.75 and the curvature 0.125.	**Rivals.** Rival kernels differ by about the ±0.03–0.06 SPARC precision. **Counts against.** Marginal today. *kernel_fingerprint*	K3
no date *needs measured σ_z*	**Edge-on disc thickness and HI flaring** Round rule against the phantom disc	σ_z 20–75% lower and HI 25–90% thicker than the phantom disc (×1.6–3.6).	**Rivals.** Standard MOND phantom disc. **Counts against.** Not diagnostic yet: the outer dispersion decides (about 7 vs 9 km/s) and no qualifying measurement exists. *CFG574–578*	K3
no date *DES ΔΣ not public*	**Drained shell and splashback shift around clusters** Engine not converged (×2 from 256^3 to 512^3)	Shell about 30% deep at 0.85–2 r_{ta}; lensing splashback shift +17–28% at 512^3 (+7–9% at 256^3).	**Rivals.** A ΛCDM splashback profile can absorb it (correlation -0.8). **Counts against.** Best forecast Z ≈ 2.1 (DES-Y3 redMaPPer); a measured lensing r_{sp} ratio below 1.07. Nothing on disk can test it. *CFG546, CFG551, CFG552*	K3
no date *resolved TDG kinematics*	**Tidal dwarf galaxies** Ownership reading (Arm C): Newtonian	NGC 5557-E1 old TDG: Newton 16–24, law + EFE 28–41, isolated law 45–58 km/s.	**Rivals.** The long-memory reading gives M_{dyn}/M_{bar} 1.4–1.9 [MI-ARM]. The two readings are not pooled. **Counts against.** Non-discriminating today (no data). *CFG441/448; tdg_test*	K3

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
no date *HI and lensing data*	**Smaller null and shape predictions** Paper-branch statements; several are contingent on the scalar being non-propagating	RAR residual blind to halo concentration (Spearman +0.012, N = 155); no starless subhalos at 10^7–10^9 M$\odot$; phantom ceiling 0.6476 a_0 at g_{bar} = 2.54 a_0; dSph σ rises about 1.9× over 40–250 kpc; red-minus-blue outer slope +0.155 / +0.128.	**Rivals.** ΛCDM assembly bias; unbounded halos. **Counts against.** Any nonzero concentration correlation; a starless perturber; a galaxy above the ceiling. The red-blue test is capped at 1.9σ. *NF P1/P4/P17–21; CFG53*	K3
no date *khronon reading*	**a_0 tied to CMB-frame speed** CFG182/186	β = 0.11–0.27.	**Rivals.** SPARC 95% limit $\beta < 2.8$; sky dipole A = 0.237, p = 0.77 (null). **Counts against.** Neither confirmed nor excluded; the conditional bound is 10–26× short. *CFG182, CFG186*	K3
no date *deep voids*	**Timescape / void reading** c-common reading	a_0 higher in voids by +0.176 dex; slope -0.145 (δ_v = -0.8) per dex of 1 + δ.	**Rivals.** Constant a_0. **Counts against.** About 266 deep-void galaxies needed; 2M++ N = 52 gives -0.083 ± 0.135, 2MRS N = 38 gives +0.029 ± 0.075. *CFG252*	K3
MI-ARM *arm closed 2026-08-08*	**Retired modified-inertia tests** Listed for completeness; they depend on the retired arm	$\eta(\beta)$ slide $d\ln\eta/d\beta$ = +0.75; infall σ-spread 0.2–1% (the old 6–13% is retracted); anisotropy-offset slope sign; SHLEM exact null window 0.43 ms (can) / 0.53 ms (alt).	**Rivals.** MG: flat or zero. **Counts against.** $\eta(\beta)$: 9 CLASH-VLT clusters give -0.09 ± 0.20; 4MOST CHANCES about 2029–31. Treat as non-live. *prep_2026 sigma_spread, cluster_efe_*, shlem_null*	K3

Structure growth and cosmology

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
no date *DESI final full-shape*	**Hubble-kernel growth branch (paper branch)** Standing record: growth fails on the gravitating field; see confronted	σ_8 +1.1% (1.5% alt); S8 0.843; fσ_8 +2.7 / 1.9 / 1.1% at z = 0.3 / 0.6 / 1.0.	**Rivals.** ΛCDM growth. **Counts against.** Low S8 or fσ_8 at the few-percent level. *PAPER15/16*	K3
no date *varied small-scale power*	**Small-scale power deficit** 512^3 boxes, near the mesh limit	P/P_{S0} 16–20% below at k = 2–3 h/Mpc.	**Rivals.** S0 (ΛCDM dark-matter-only PM). **Counts against.** Not converged; an engine artefact in part (CFG526). *CFG521/526/527*	K3
DESI Lyα *date not on record*	**Small-scale cutoff** Lyman-α forest	Half-mode k ≈ 35 h/Mpc; Murgia $\alpha < 0.03$ passes (derived α = 0.0190).	**Rivals.** CDM. **Counts against.** A cutoff mismatch. Today 0.4–0.9σ (the '6–8σ' was withdrawn). *SWEEP3; forest_bcut*	K3
2030s *LISA / Taiji / TianQin, DECIGO*	**Primordial-black-hole window** Asteroid-mass window as the cold energy; literature bounds recalled, unverified	Window 10^{17}–10^{22} g; induced GW peak 1.2 mHz (10^{22} g) to 0.36 Hz (10^{17} g), $\Omega_{GW}h^2$ about 4–5e-9.	**Rivals.** No PBH cold energy. **Counts against.** A null LISA background rules out Gaussian adiabatic formation across most of the window. The amount is a restatement of Ω_c. *CFG510*	K3
2030s *LISA, ET*	**Gravitational waves and EMRI dress** c_T = c and α_M = 0	Relaxed phantom: EMRI phase $\Delta\Phi$ 5e-8 to 5e-6 rad (undetectable). Unrelaxed spike 11 to 1.2e4 rad. $\Xi \equiv 1$.	**Rivals.** Spike-level dress; $\alpha_M \neq 0$. **Counts against.** $\Delta\Phi \gtrsim$ 0.1–1 rad kills the relaxed rule; a confirmed $\alpha_M \neq 0$ falsifies the class. A null does not discriminate. Superradiance leaves wave-field mass windows 2.0–4.4e-20 eV and 5.3e-17 to 6.8e-13 eV. *CFG512, CFG367*	K3
no date *binary-pulsar timing*	**Khronon dipole radiation** Conditional pass	α_c in [9.6e-14, 3.2e-9]; largest $\dot{P}_b$ deviation 4.9e-13 (wave-zone) or 1.9e-8 (universal bound). Flux-ledger repair is period-independent.	**Rivals.** GR. **Counts against.** A period-dependent deviation. *CFG291*	K3
no date *direct detection*	**No dark-matter particle** A requirement	No direct or indirect detection; DM cannot decay or annihilate; no 11–28 eV thermal relic. Early stiff component $\Omega \lesssim$ 4e-25 (a 24-order fine tuning, not a kill).	**Rivals.** Particle dark matter. **Counts against.** A confirmed detection. *NF P6/P10/P16*	K3

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
BepiColombo MORE *arrival Nov 2026 per one inventory*	**Solar-system residuals** Unverified date	Any MOND detection would exclude the F(Z) class; a coherent apsidal residual would revive F(Y). Planetary corner needs ω_c in [1.78e-14, 2.21e-14] rad/s (can).	**Rivals.** GR / ephemerides. **Counts against.** Reading B (a_0/c drift) already excluded ×246 (MESSENGER) and 47–57σ (LLR). The per-body solar-reflex reading (0.4 cm–1.2 m against MORE's ~1 cm; orbit insertion 21 Nov 2026 per one note) is superseded. Saturn: an anomalous sunward acceleration of 5.84e-16 / 8.53e-16 m/s^2 (can / alt) from the n = 2 power-law kernel, about 17× below Cassini's residual; a null ranging below that kills the kernel (branch row). *ET:62-63; planetary_doors*	K3

Neutrino and particle sector

All conditional on tri-maximal TM1 mixing, Dirac neutrinos and $m_1 = {\rho_\Lambda}^{1/4}$. A coherent chain is not evidence. The particle-numerology search found no certified interlock (42,534,139 distinct, 0 certified).

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
2027–28 *JUNO*	**Solar mixing angle** KILL_SHEET A1	$\sin^2\theta_{12}$ in [0.3162, 0.3198] for ${s_{13}}^2$ in [0.020, 0.025] (central today 0.309).	**Rivals.** Other textures; TM2 and golden ratio sit 3.6–3.8σ off. **Counts against.** The whole 3σ range outside the window. *KILL_SHEET:27; atomos FS16*	K3
~2030 *JUNO*	**Mass ordering** A5	Normal ordering.	**Rivals.** Inverted ordering. **Counts against.** Inverted ordering established. *KILL_SHEET:31*	K3
2030s *NOvA/T2K, DUNE, Hyper-K*	**CP phase and octant** A2	sign(cos δ) = sign(${s_{23}}^2$ - $\frac{1}{2}$); $\cos^2\delta \leq 0.21$; $\lvert J\rvert \geq 0.0277$.	**Rivals.** Other δ. **Counts against.** δ near 0° or 180°. *KILL_SHEET:28*	K3
2030s *LEGEND-1000, nEXO*	**No neutrinoless double beta decay** A3 (Dirac)	$m_{\beta\beta} = 0$. Dirac needs $m_{lightest}$ below 7.66 meV (NO).	**Rivals.** Majorana. **Counts against.** Any $0\nu\beta\beta$ signal. *KILL_SHEET:29; FS11*	K3
~2030 *DESI DR3 + Euclid + CMB-S4*	**Sum of neutrino masses** A4	Σm_ν = 0.0613–0.0614 eV (masses 2.24 / 8.95 / 50.3 meV); $m_\beta \approx 9$ meV (unreachable by KATRIN, about 22× below).	**Rivals.** Minimal normal-ordering floor 0.0587 eV, 0.06σ away. **Counts against.** A 95% bound below 0.0609 eV. It already shows about 3σ effective-mass tension with DESI, shared with the minimal ordering. Growing-mass tower: $m_\nu(z{=}3)/m_\nu(0)$ = 0.59–0.75, dies if w = -1. *KILL_SHEET:30; CFG507; neutrino_desi*	K3
no date *CMB B-modes*	**Inflation by stochastic misalignment** CFG362	$H_I \leq 83$ GeV.	**Rivals.** High-scale inflation. **Counts against.** A B-mode detection. *CFG362*	K3

Added in the second sweep: lensing, clusters, kernels

Rows from real_research, hunt_2026 and the paper deposits. CANDIDATE-CONDITIONAL rows come from the old clock-host candidate (PAPER32 says it fails cosmology as built).

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
Euclid shear, mid-2027+ *Euclid LE3; DR1 images Nov 2026*	**Weak-lensing RAR deep tail (Data-gate 1)** Pre-registered 2026-06-26 (DOI 10.5281/zenodo.20935948); survives only as a MOND-class test	Deep slope 0.5000. The a_0 value is only a 0.054 dex shift (9.36e-11 vs 1.2e-10), below the 0.10 dex baryonic systematic.	**Rivals.** Any MOND-class law shares the slope; ΛCDM halos steepen. **Counts against.** Slope off 0.5 at $> 3\sigma$, or $g_{obs} \to g_{bar}$. Expected: CONFIRM on the shared front, INCONCLUSIVE on the a_0 value. *DATA_GATE_PREREGISTRATION_2026.md*	K2
no date *stacked cluster shear beyond R500*	**Cluster outer weak-lensing slope (COS1)** Frozen; the registered check C5 failed, so INCONCLUSIVE as registered	ESD slope over [R500, 3R500]: -0.997 (alt -0.998); $M_{tot}(<R500) = 3.93\ M_b$, outside the [1.65, 2.0] anchor band.	**Rivals.** NFW: -1.591 to -1.673 (contrast +0.642). **Counts against.** An NFW-like slope. About 4.3e3 stacked lenses for 3σ; the data gate is not met. May overlap the Euclid kink row. *real_research/cluster_outer_slope_2026*	K2
no date *KiDS / Euclid isolated-lens stacks*	**Lensing truncation of boosted halos** Exploratory lead, $\Delta\chi^2$ -15 to -19	$r_t = v_{flat}/(\sqrt{x_c}\ H(z))$: about 1.1 Mpc for the Milky Way today (x_c about 4–7).	**Rivals.** No truncation. **Counts against.** No truncation, or $x_c \geq 15$. *THE_THEORY_AS_IT_STANDS_2026-09-22.md*	K3
no date *low-z Lyα (HST/COS), tSZ stacks*	**Active-filament gas at z < 1** The switch turns the law on where $\delta \geq 3.7$–6.3 (uncomputed)	Force ×3.5–10 in filaments; infall 3.2–5.9 Gyr against 7.9 Gyr since z = 1.	**Rivals.** Newtonian filament y and b–N. **Counts against.** A Newtonian filament signal. *cross_thread_review XR4*	K3
on disk, not run *KiDS Fig. A4*	**All-galaxy against isolated lensing split** Brouwer+21	The construction must reproduce the measured split of +0.40 dex (+152%) at the lowest g_{bar}.	**Rivals.** — **Counts against.** Failure to reproduce. *XR4 gates*	K3
no date *~20 clusters at z ≈ 1, one pipeline (HSC, CLASH/RELICS)*	**Halo concentration evolution (the crispy gap)** ΛCDM's side of the flat-a_0 test	Flat a_0 needs no dilution of halo concentration with redshift.	**Rivals.** To mimic a flat a_0, ΛCDM halos at fixed mass must dilute as $c/c_{Nbody} \approx (H/H_0)^{-2/3}$: 0.61 at z = 2, 0.40 at z = 3, 0.12 at z = 5. NFW a_s grows ×1.08 / 1.23 / 1.76 / 2.57 at z = 0.5 / 1 / 2 / 3. **Counts against.** Concentrations at N-body values at $z \geq 2$ while a_0 stays flat means the framework's a_0 is not emergent; diluted concentrations let ΛCDM survive a flat a_0. About 20 clusters for 3σ; no homogeneous sample today. *ledger A-9; SLH; crispy_2026*	K3
no date *ultra-deep isolated rotators (BIG-SPARC)*	**Deep-MOND floor** Floor against external-field fork; AGC 114905 has an inclination dispute	A universal downturn at a* below 0.107 a_0 (95%, SPARC); target $e_N \lesssim 0.005$ and g_{obs} 0.01–0.05 a_0.	**Rivals.** No universal downturn if the external field acts. **Counts against.** No downturn. *WATCHLIST.md*	K3
no date *age-dated isolated dwarfs*	**Young isolated rotators: transient fingerprint** Three-way split by settling state	IMPL +0.3 to +1.9 dex above the RAR; COMP quasi-Newtonian; settled on the RAR. TDG gap 0.21 dex, N ≈ 12 TDGs for 3σ.	**Rivals.** TDG data -0.13 ± 0.10, non-diagnostic. **Counts against.** Depends on the branch. *WATCHLIST.md*	K3
no date *deep ALMA [CII] to 3–4 R_e*	**REBELS-25 high-z fork** Declining, constant and rising a_0 split the asymptotic velocity ×2.3–2.4	Declining 86–143, constant 106–175, rising 204–335 km/s (can).	**Rivals.** — **Counts against.** Needs M_{bar} to ×2 (currently ×7). *WATCHLIST.md*	K3
no date *precision RAR / BTFR at g_{bar} 0.05–0.4 a_0*	**Kernel fingerprint: finite-acceleration BTFR lift** Exponential kernel; point-mass limit only	$v_c^4/(G\ M\ a_0) = 1 + \frac{1}{2}\sqrt{s} + (5/24)s$: +16% at 0.1 a_0, +34% at 0.3 a_0. Exterior-curve 1/r correction $v^2 = \sqrt{(GMa_0)} + GM/4r + 7(GM)^{3/2}/(96\ r^2\ \sqrt{a_0})$.	**Rivals.** Simple kernel $c_1 = 1$; standard kernel 0. **Counts against.** SPARC smears the lift; the exponential kernel ranks just below the RAR kernel. Extended mass and EFE dilute the 1/r term. *FINDINGS L92/L109; EXACT_EXPONENTIAL ledger*	K3
no date *isolated low-g host with two tracers*	**Deep-MOND action-angle invariant** Static test-particle result	$T_r\sqrt{(GMa_0)}/\ell = I(e)$, $I(e\to 0) = \pi\sqrt{2} = 4.4429$: two equal-e tracers of one host give a period × angular-momentum ratio of 1 (Newton: a_1/a_2).	**Rivals.** Newtonian ratio. **Counts against.** EFE, finite g/a_0 and non-spherical baryons deform I(e). *FINDINGS L233*	K3

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
runnable now *Gaia orbits + archival spectra*	**Dwarf dispersion against Galactocentric radius (EFE)** CANDIDATE-CONDITIONAL; the Local Group dwarf EFE slope already reads +0.0800 ± 0.0467 against a corrected -0.093	σ rises ×1.95 / ×1.88 over R_{gc} 40–250 kpc; 17 named dwarfs give about 3.9σ; about 10 dwarfs for 3σ, about 28 for 5σ.	**Rivals.** DM predicts flat. **Counts against.** Tidal stripping gives a same-sign correlation in ΛCDM (needs σ against pericentre at fixed R_{gc}); the McConnachie dwarfs gave a null. *FINDINGS L25; THE_LIABILITY_TABLE*	K3
no date *KiDS / Euclid shear stacked on 2M++ g_{ext}*	**Lensing quadrupole aligned with the external field** Not runnable: no shear catalogue on disk	cos 2ϕ modulation of $\Delta\Sigma$ about 0.21 of the monopole at 0.5 Mpc (e_N = 0.01), S/N ≈ 3.8 on KiDS-1000 errors.	**Rivals.** NFW q = 0.8: 0.048. **Counts against.** No modulation. *hunt_2026/h83*	K3
no date *galaxy-galaxy lensing binned by acceleration*	**Lensing-versus-dynamics slip** CANDIDATE-CONDITIONAL	F(Q)Θ carrier: no slip. A QUMOND-type carrier slips ∝ (ν - 2), zero at g_{bar}/a_0 = ln 2 = 0.693.	**Rivals.** — **Counts against.** Needs the coupled (Φ, Ψ) profile solve; a later lane gives 4e-7 for the candidate. *FINDINGS L279-280*	K3
no date *EDGES / SARAS-3*	**Collapse-timing cost for the 21-cm onset** A cost, not a linear signature (the 'not predicted' list still holds for linear 21-cm)	Collapse speed-up 1.24–1.4× at z = 17–25 advances first stars: onset earlier by Δz 1–2; τ_e 0.0700–0.0703 (+2.1 / +2.2σ).	**Rivals.** ΛCDM onset. **Counts against.** Needs an ionizing efficiency below ΛCDM's by a definite factor; JWST $\xi_{ion}f_{esc}$ is uncertain ×2–3. *SLH:162; hunt_2026/h73*	K3
no date *kSZ + independent τ (FRB DM)*	**kSZ pairwise velocity** Untestable today	v_{12} +2.12% (can) / +2.97% (alt) at z = 0.5 (192 km/s at 20 h^{-1} Mpc).	**Rivals.** ΛCDM. **Counts against.** Best case 0.30σ; needs σ_τ/τ below 0.71% against 10% best. *FINDINGS.md:7051-7059*	K3

Added: wide-binary readings from other kernels (do not pool)

These are alternative-kernel readings tagged BRANCH or CANDIDATE-CONDITIONAL. They are not the registered arms, and the registered arms are the only ones DR4 scores.

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
2026-12-02 *Gaia DR4*	**Alternative-kernel wide-binary values** BRANCH / CANDIDATE-CONDITIONAL; conflicts flagged	F(Q)Θ exponential kernel: γ_v 1.033 can / 1.059 alt (anisotropy 1.08–1.12 parallel, 1.01–1.03 perpendicular). Photocount-EFE reading: 1.0953 can / 1.1240 alt (n = 2: 1.095–1.111). PAPER11: 1.03–1.06. PAPER44 EFE-blind: 1.20 at 10 kAU, 1.51 at 20 kAU.	**Rivals.** Registered Arm A band 1.16–1.23. DeepSeek equilibrium branch: γ_v band [1.047, 1.11], P–s excess +18.4%, cap break at 7.4 kAU. **Counts against.** The exponential-kernel anisotropy sign (parallel > perpendicular) is opposite to the frozen pre-registration. PAPER44 was not deposited and CFG447 says Gaia DR3 excludes it. *FINDINGS L25; kimik3 predictions; PAPER11/44; deepseek DR4_AMENDMENT_12*	K3

Added: settling-clock and cluster predictions (AI branch)

BRANCH: from the OpenAI math cross-analysis and DeepSeek folders. They are non-authoritative, and some conflict with the standing record. The lane's own cold-budget audit (T15) counts against the framework (see confronted).

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
no date *MW halo tracers at 60–200 kpc*	**Settling front and outer decline (T17)** BRANCH; not in the earlier record	V(200 kpc)/V(60 kpc) = 0.63; c = 55–59 kpc; r_f = 80.0 (V = 188) / 85.1 (V = 200) kpc; $M_{dark}(\infty)$ = 0.96–1.22 × 5.36 M_b.	**Rivals.** Flat curve at 150–200 kpc. **Counts against.** V(200)/V(60) above 0.85; outer slope not below about 0.3 by 2 r_f. *OMX t17*	K3
no date *epoch-split completeness stacks*	**Completeness elasticity and scatter (T12, T14)** BRANCH; partly self-undercut by T15 (the f-values were leftover fractions)	$\epsilon(f) = (1-f)(-\ln(1-f))/f$: $\epsilon(0.14)$ 0.9265, $\epsilon(0.60)$ 0.6109, $\epsilon(0.43)$ 0.7451; $\sigma(f)/f = \epsilon\ \sigma(\ln t_f)$, ratio 1.146 between clusters and groups.	**Rivals.** Flat ϵ; constant scatter. **Counts against.** A flat ϵ kills it; scatter outside 0.4–0.5 by more than 2σ. Needs about 30% per bin. *OMX t12, t14, t15*	K3
no date *BCG / red-sequence archaeology*	**Assembly clock (T11)** BRANCH; the literature already lands inside the window	$t_c/t_g = \ln(1-f_c)/\ln(1-f_g)$ = 0.6135, window [0.55, 0.66]; no mass trend at fixed Δ.	**Rivals.** — **Counts against.** A ratio outside the window. *OMX t11*	K3
no date *SPARC residual structure*	**Settling-flattening (T10)** BRANCH	Bump relaxation $\tau \propto \ell^2$ (measured exponent 1.986); relaxation 10^8–10^9 yr below about 10 kpc.	**Rivals.** White-noise residuals. **Counts against.** No radial correlation in residuals. *OMX t10*	K3
no date *eROSITA eRASS3 cluster catalogue*	**Cluster redshift invariance (G237)** BRANCH; supports flat a_0 against $a_0 \propto H(z)$; catalogue-only, pending	u = r_M/R500 = 0.1850 at every z below 3; cluster pie $s_b/s_{ph}/s_d$ = 0.177/0.569/0.254; ΔlogT = 0.0000 dex.	**Rivals.** $a_0 \propto H(z)$: +0.0996 dex (about 9σ with 50-cluster bins). **Counts against.** |ΔlogT| above 0.010 dex at 2σ in any z-bin 0.2–1.0; |s_b - 0.177| above 0.020; u(z)/u(0) off by more than 10%. *deepseek G237*	K3
2026-12-02 *Gaia DR4 vertical Jeans*	**Vertical funnel and universal dark column** BRANCH; may overlap the sealed CFG553 row	z_c(8.2 kpc) = 140.63 pc (can) / 169.42 pc (alt), flare $e^{R/3}$; $\Sigma_{ph}(<r_M) = a_0/(\pi G)$ = 213.74 $M\odot/pc^2$.	**Rivals.** NFW inner column 6.6 against 27.8. **Counts against.** Flat or random z_c(R); a positive dark mass at |z| 300–560 pc kills the slab. *deepseek DR4_AMENDMENT_12; G237*	K3
no date *stacked tSZ, X-COP, XRISM*	**Cluster inner slope, tSZ outer slope and temperature plateau** BRANCH	Inner mass slope about -1.5 (measured -1.53 ± 0.25; -1 excluded at 7.2σ); outer Compton slope -2.54 (-2.7 to -2.05) against phantom-only -1.44; T plateau median 3.63 keV.	**Rivals.** NFW -1. **Counts against.** Inner slope -1 kills the two-component build. *deepseek FALSIFIER_MATRIX 5–9*	K3
no date *SDSS-class close pairs, ΛCDM mock control*	**Merger-pair excess** BRANCH	$f_{pair}/f_{\Lambda CDM} = (1 + r_M/s)^q$, q = 1: 2.0 at s = r_M, 1.5 at $2r_M$, 3.0 at $r_M/2$.	**Rivals.** ΛCDM mock. **Counts against.** Anchor ≤ 1.00; q outside [0.5, 1.5]; scatter above 20%. *deepseek FALSIFIER_MATRIX 4*	K3
~1 year *H3/APOGEE + GD-1/Pal 5/Orphan*	**Milky Way break width** BRANCH; adjacent to the Euclid kink	First-order step w90 below 0.3 kpc, r_{peak} 6.33 kpc, step +0.221 at 3.07σ.	**Rivals.** Smooth crossover 1–2 kpc. **Counts against.** w90 - 3σ above 2 kpc falsifies both readings. *deepseek FALSIFIER_MATRIX 14*	K3
no date *RAR samples at high z*	**RAR scatter growing with redshift (H025)** BRANCH, frozen 2026-09-15, no result; rests on growing external field, in tension with flat a_0	$\sigma(z) = \sigma_0$ s(0)/s(z): 0.064 / 0.091 / 0.122 / 0.168 / 0.229 / 0.293 / 0.44 dex at z = 0 / 0.3 / 0.6 / 1 / 1.5 / 2 / 3.	**Rivals.** Constant scatter. **Counts against.** Pass needs ≥ 1.5× the local floor; fail if ≤ 1.2× in every bin. *hy4_push H025*	K3
no date *next-generation ranging*	**Saturn anomalous acceleration (power-law kernel)** BRANCH (kimik3); the exponential kernel predicts zero	5.84e-16 (can) / 8.53e-16 (alt) m/s^2, about 17× below Cassini's residual.	**Rivals.** Zero (exponential kernel); Newton. **Counts against.** Ranging below 8.5e-16 with a null kills the n = 2 kernel; a detection kills the exponential kernel. *kimik3 predictions P5/N1*	K3
no date *SPARC rebinned in baryonic mass*	**Universal retained dark fraction (C003)** CANDIDATE-CONDITIONAL; not derived from an action	Retained fraction 0.128 spiral, 0.132 MW, 0.168 group, 0.664 cluster at v_k = 700 (can); 0.127 / 0.132 / 0.155 / 0.494 at 1000 (alt).	**Rivals.** — **Counts against.** Trend steeper than 1.2× across 10^9–10^{11} M⊙; the spiral value is 22% over the ceiling. *FINDINGS L295-312; kimik3 P6*	K3
no date *DESI / Euclid growth*	**Growth step from growing neutrino mass** Conditional on the neutrino chain	Scale-dependent step 0.65% at Σ = 0.059 eV, 1.39% at 0.10 eV (1.4σ at the floor, 3.9σ at 0.10 eV); Σ_{eff} 0.032–0.042 eV.	**Rivals.** Scale-independent dark-energy growth. **Counts against.** w → -1, or Σ below 0.059 eV. *GROWING_NU_GROWTH_TOMOGRAPHY_2026.md*	K3

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
no date *$z \approx 1.5$ discs with measured CO*	**Gas demand of KURVS-like discs** KILL_SHEET G7	Flat a_0 needs gas ≈ 4 M*.	**Rivals.** — **Counts against.** A measured gas-to-star ratio $\mu \leq 1.5$. *KILL_SHEET G7*	K3

4 Neutrino predictions in detail

Everything the record says about neutrinos in one place. Each number is read from the file named in the source line.

What is forced and what is not. The framework forces one number: the vacuum-energy scale $E_L = {\rho_{DE}}^{1/4} = 2.2395$ meV. It does **not** force $m_{lightest} = E_L$; setting the O(1) coefficient to exactly 1 is a fit by inversion, with no geometric factor behind it. Any cosmic density$^{1/4}$ (critical, matter, 10× dark energy) lands at 1.3–3.6 meV, so the match to neutrino masses is generic, not specific to ρ_{DE}. Seven literature mechanisms were computed and none sets m_ν (the MaVaN acceleron is the one real candidate, but its coupling scale is fitted and it carries a ${c_s}^2 < 0$ instability). Falsifying this hypothesis leaves $a_0 = \sqrt{\Lambda}$ untouched.

The predicted spectrum (normal ordering, $m_1 = E_L$)

Quantity	Canonical	Alt footing	Current bound or reach	Reading
m_1	2.2395 meV	2.46 meV	DESI caps m_1 below 5.0 meV at 3σ	E_L forced; $m_1 = E_L$ not forced
$m_2 = \sqrt{({m_1}^2 + \Delta {m^2}_{21})}$	8.900 meV	9.00 meV	$\Delta {m^2}_{21} = (7.42 \pm 0.205)$e-5 eV^2	derived from oscillation data
$m_3 = \sqrt{({m_1}^2 + \Delta {m^2}_{31})}$	50.15 meV	50.2–50.4 meV	$\Delta {m^2}_{31} = (2.510 \pm 0.027)$e-3 eV^2	derived from oscillation data
Σm_ν	**0.0613 eV** (0.06129)	0.0617–0.0619 eV	DESI+CMB (ΛCDM) below 0.0642 eV; Feldman–Cousins below 0.053; $w_0 w_a$ below 0.163	2.58 meV above the minimal normal-ordering floor 0.0587 eV (0.06σ): indistinguishable from it
m_β (tritium)	9.07 meV (0.00907 eV)	about 9.2 meV	KATRIN below 0.45 eV now, about 0.2 eV final; Project 8 target about 0.04 eV	22× below KATRIN-final: not a test this decade
$m_{\beta\beta}$ ($0\nu\beta\beta$, Majorana case)	0.006 to 5.27 meV over the phases	—	28–122 meV now; 5–20 meV next generation (nEXO, LEGEND-1000)	at or below reach; a non-detection cannot kill it. In the Dirac reading $m_{\beta\beta} = 0$ exactly
Inverted-ordering control ($m_3 = E_L$)	$\Sigma = 0.1014$ eV	—	excluded at DESI's 95% ΛCDM bound	m_β 48.9 meV; $m_{\beta\beta}$ 18.7–48.4 meV (a guaranteed $0\nu\beta\beta$ signal)

Two input sets. The canonical column uses the June note (NuFIT 5.3 / PDG 2024: $\Delta {m^2}_{21}$ 7.42e-5, $\Delta {m^2}_{31}$ 2.510e-3 eV^2). The atomos lane FS16 uses the 2025 table (NuFIT 6.0 plus JUNO) and gives, canonical: m_1 2.24, m_2 8.95, m_3 50.2–50.4 meV, Σ 0.0614–0.0616 eV, m_β about 9.2 meV; alt: 2.46 / 9.00 / 50.2–50.4, Σ 0.0617–0.0619 eV. The kill sheet quotes $\Sigma \approx 0.0614$. The alt column above is FS16's. The differences are input choices, not different predictions. The inverted-ordering control is 0.1014 eV in the June note, 102.9 meV in the posits slate, and only bounded below at 0.0970 eV (Lean L11); all agree it is excluded.

Checked against DESI DR2 on 2026-10-06 (CFG507; Elbers+2025, arXiv:2503.14744)

Check	Result
$m_{lightest}$ below 0.023 eV (95%)	PASS (2.2 meV)
Σ below 0.0642 eV (ΛCDM, 95%)	PASS by 0.0029 eV (0.15σ)
Σ below 0.053 eV (Feldman–Cousins, 95%)	FAILS
Σ below 0.163 eV (w_0w_a, 95%)	PASS
Effective-mass tension	FAILS at about 3.0σ, the same tension DESI + CMB in ΛCDM puts on every mass the oscillation data allow. It gets no worse. Never call this support.
MUTATE (plant the inverted floor 0.101 eV)	the ΛCDM limit flips to FAIL (4/7 checks); main run 5/7

The test chain, soonest first

When	Facility	Prediction	Counts against it	Grade
2026–28	DESI DR3 + CMB	$\Sigma m_\nu = 0.0613$ eV (canonical), at the edge of a narrowing band	A robust 95% bound at or below about 0.060 eV (the file's own sharpest falsifier). Cannot separate it from minimal normal ordering.	K2
~2027–28	JUNO	TM1: $\sin^2\theta_{12}$ in [0.3162, 0.3198] for ${s_{13}}^2$ in [0.020, 0.025]. JUNO 2025 gives 0.3092 ± 0.0087; combined 0.3084 ± 0.0069; the chain's central 0.318 is $+1.4\sigma$. TM2 and golden ratio are already 3.6 / 3.8σ off.	JUNO's whole 3σ range outside the window.	K2
mid-2027 to ~2030	JUNO, others	Normal ordering	Inverted ordering established (the only viable reading is normal).	K2
ongoing to 2030s	NOvA / T2K, DUNE, Hyper-K	TM1: $\text{sign}(\cos\delta) = \text{sign}({s_{23}}^2 - \frac{1}{2})$; $\cos^2\delta \leq 0.21$ (δ within 27.3° of 90° or 270°); $\lvert J\rvert \geq 0.0277$. Lower octant gives δ about 262° (coherent, 1.9σ with NuFIT 6.0 IC24: s_{23} 0.470, δ 212 +26/-41); upper octant gives about 285° (in tension; IC19 s_{23} 0.561, δ 177 +19/-20). The octant decides.	δ near 0° or 180°, or the octant and the sign of cos δ disagree.	K3
2030s	LEGEND-1000, nEXO	$m_{\beta\beta} = 0$ (Dirac). The AdS-swampland argument forbids Majorana neutrinos and allows Dirac for $m_{lightest}$ below 7.66 meV (NO) or 2.46 meV (IO).	Any $0\nu\beta\beta$ signal (also closes the Dirac window).	K3
~2030	DESI + Euclid + CMB-S4	$\Sigma m_\nu = 0.0614$ eV. If the lightest neutrino is a swampland-tower state, $m_\nu(z=3)/m_\nu(0) = 0.59$–0.75 (DESY5 0.65, Union3 0.59, Pantheon+ 0.75; naive ${\rho_{DE}}^{1/4}$ law 0.78–0.84). Constant-mass CMB fits would then read $\Sigma_{eff} =$ 0.032–0.042 eV.	A 95% bound below 0.0609 eV; w $\to$ -1 (no roll means no mass evolution). The mass shift in Σ itself is tiny (about 0.15 meV); the relaxation is the generic w_0w_a degeneracy.	K3

When	Facility	Prediction	Counts against it	Grade
2026–28	Euclid, DESI growth	A scale-dependent growth step from a growing neutrino mass: 0.65% at $\Sigma = 0.059$ eV, 1.39% at 0.10 eV. Euclid Fisher: 1.4σ at the floor, 3.9σ at 0.10 eV.	No step; or w → -1. Dynamical dark energy is scale-independent, which is the separator.	K3
no date	CMB B-modes	Inflation by stochastic misalignment: $H_I \leq 83$ GeV.	A B-mode detection.	K3

Bounds and implications certified in Lean (LEPTON_CHAIN 1–3, MODULAR_S4_FIXED)

Zero sorry, each with a mutation that must fail. These certify implications from stated assumptions, not nature. The assumptions (TM1, Dirac, $m_1 = \rho_\Lambda{}^{1/4}$, $\kappa = \frac{1}{2}$) are not certified.

Label	Statement	Number	Kind
L2	TM1 fixes $\sin^2\theta_{12}$ for ${s_{13}}^2$ in [0.020, 0.025]	[0.3162, 0.3198]	kill line: a 3σ ${s_{12}}^2$ bound outside it excludes TM1
L3	TM2 gives ${s_{12}}^2 \geq 1/3$	3–4σ off JUNO	kill result for TM2
L4	$\text{sign}(\cos\delta) = \text{sign}({s_{23}}^2 - \frac{1}{2})$ for ${s_{13}}^2 < 1/5$; maximal θ_{23} gives $\cos\delta = 0$	—	a measured octant and sign mismatch kills TM1
L5	m_1 = 2.24e-3 eV with NuFIT-range splittings gives Σ in	0.0609 to 0.0620 eV	below DESI DR2's 0.0642
L6	TM1 with ${s_{13}}^2 \leq 0.0228$ gives m_β	< 0.0095 eV	below Project 8's goal: untestable
L7	If $a_0 = \kappa c\sqrt{(G\rho)}$ and $m^4 = K\rho$ then $m_{\text{lightest}} \propto \sqrt{a_0}$ at fixed κ	footing ratio 1.21	algebra; rests on assumption A3
L8	Dirac neutrinos give $m_{\beta\beta} = 0$	0	definitional, not physics content
L9, L10	TM1 forbids CP conservation: $\cos^2\delta \leq 0.21$, $\lvert J\rvert \geq 0.0277$	scan max $\cos^2\delta$ 0.2061; min $\lvert J\rvert$ 0.02826; box max 0.0358	CP violation at least 77% of maximal; δ within 27.3° of 90° or 270°
L11	Inverted ordering with $\lvert\Delta{m^2}_{32}\rvert \geq$ 2.40e-3 gives Σ above 0.0970 eV	0.0970 (control 0.1014)	excluded at DESI's 95% ΛCDM bound; never "DESI forces normal ordering"
L12	Normal ordering with $\Sigma <$ 0.0642 eV gives m_1	< 4.6 meV (mutation: 4.2 meV gives Σ 0.06384)	a data bound, tighter than the 7.35–7.66 meV Dirac window
L13	Under L12 and L7, a_0 is capped	$a_0 <$ 4.22 × canonical (Lean file) against 4.99 × (kill sheet A7)	weak; the two documents disagree and it is unresolved. Both footings pass
S5, S6	Full Klein symmetry gives $\theta_{13} = 0$ (excluded); the transposition Z_2(SU) alone gives TM1; S alone gives TM2	—	the choice of residual Z_2 is a hypothesis; modular fixed points reach TM1 but not TM2 (24/48 coin flip)

Mixing-angle and CP numbers in more detail (FS05, FS08, FS14–FS16)

Quantity	Value	Against the data
TM1 $\sin^2\theta_{12}$ (chain central)	0.3183	$+1.4\sigma$ against the JUNO × NuFIT 6.0 combination (0.3084 ± 0.0069); JUNO alone 0.3092 ± 0.0087
θ_{12} sum-rule pulls	TM1 0.94 / 0.86 (IC19 / IC24), 1.05 JUNO alone	TM2 2.82 / 2.74 / 3.64; golden ratio -2.78 / -2.87 / -3.77
TM1 δ, no octant input (FS05)	289° ± 5° or 71° ± 5°; $\lvert J\rvert$ = 0.03171 (refit 288.8 / 70.7)	null p_δ = 0.23 (362 of 1550 random columns compatible): not discriminating against chance

Quantity	Value	Against the data
IC19 (no SK), upper octant, s_{23} 0.561, δ 177 +19/-20 (3σ 96–422)	$\delta = 74.2°$ or 285.4°	pull 5.05 / 5.6: DISFAVOURED
IC24 (with SK), lower octant, s_{23} 0.470, δ 212 +26/-41 (3σ 124–364)	$\delta = 97.5°$ or $262.1° \pm 3.9°$	pull 2.78 / 1.91: CONSISTENT. The octant decides which branch survives
Majorana phases	none predicted	moot under the Dirac reading
θ_{13} and θ_{23} point values	inputs only ($s_{13}{}^2$ box [0.020, 0.025]; θ_{23} via the octant rule)	no point prediction exists

Dirac window and the AdS conjecture (FS11, FS12, FS18)

Item	Result
Majorana neutrinos	Forbidden (an AdS minimum at every mass). Conditional on the unproven non-SUSY AdS swampland conjecture; the literature anchor (Hamada–Shiu) agrees on Dirac with O(1–10) meV.
Dirac window	$m_{lightest}$ below 7.66 meV (NO; refined from 7.35, next grid point 8.43 forbidden) or 2.46 meV (IO). With $\rho_\Lambda = 0$ the window closes, so Λ sets it: the only computed link from Λ to a flavour parameter, and it gives a bound, not a derivation.
Extra light bosons (FS12)	The NO window closes at 3 extra massless bosonic degrees of freedom (IO at 1); N = 7 forbids 2.24 meV under NO; N = 8 reopens (unresolved). So the framework may carry at most 2.
Evolving dark energy (FS18, not crispy)	m_1 allowed ceiling by $s = \rho_{DE}/\rho_0$: 0.057 meV at 1e-5, 1.16 at 1e-2, 7.66 at 1, 15.7 at 10. A constant 2.24 meV violates only below $s^* = 0.045$. The lowest ρ_{DE}/ρ_0 on $z \leq 3$ is 0.548 (DESY5), 0.601 (Pantheon+), 0.499 (Union3), so violations appear only at $z \approx 15$–35 under a CPL extrapolation.
Texture and modular doors	No viable Dirac two-zero texture (FS13c) and Dirac textures cannot fix the absolute mass; the one Majorana-side survivor A2 × TM1 (Σ 0.0651–0.0659 eV, m_{light} about 5.1 meV) sits above DESI and is forbidden by FS11; the minimal weight-2 modular Dirac model is excluded (χ^2 about 47,000); μ–τ reflection ($s_{23} = 0.5$, $\delta = \pm 90°$) is excluded at 3σ (s_{23} 4.0σ off 0.572).

Cosmology links: growth, dark radiation, BBN

Item	Number	Counts against it / status
CMB mass offset if the mass tracks $a_0(z)$ (growing neutrino mass)	m_ν(CMB)/m_ν(today) = 0.54–0.72 (DR1), 0.57–0.74 (DR2): 28–46% lighter; inferred Σ_{eff} 0.032–0.042 eV for true Σ = 0.059 eV	w → -1 (no roll, no offset). The DR1→DR2 shift is under 0.05, an internal stability check rather than a data pass. Zenodo 10.5281/zenodo.20977421.
Co-evolution exponent	$m_\nu \propto \rho_{DE}{}^p$ gives $m_\nu(z)/m_\nu(0) = [a_0(z)/a_0(0)]^{2p}$; $a_0(3)/a_0(0)$ = 0.61–0.71 (DR1), equal to the mass ratio only at $p = \frac{1}{2}$; 0.38–0.51 at p = 1	p is a free knob, so this is a hypothesis with a free parameter, not a forced prediction.
Shared w(z) lock	CMB-versus-late Σ split Δ_Σ = 16–27 meV against CMB-S4 $\sigma(\Sigma)$ about 16–32 meV (about 1–2σ target)	w → -1. A consequence only if the exponent link holds.
Scale-dependent growth step	0.65% (Σ = 0.059 eV) and 1.39% (0.10 eV); small-scale (k > 0.2) suppression 1.09% / 2.00%; k_{fs} 0.007–0.016 h/Mpc; Euclid Fisher 1.4σ / 3.9σ	A rough Fisher with no bias or nonlinear marginalisation; its power rests on the absolute mass that JUNO and KATRIN pin.
Dark radiation from a steeper quartic well	$\Delta N_{eff} = 0.094\ (1/\mu_\Psi$ / 10 Mpc)^2; Planck $\Delta N_{eff} < 0.3$ allows $1/\mu_\Psi$ up to 18 Mpc; the lensing RAR wants above about 1–3 Mpc	CMB-S4 (±0.06) decides above 8 Mpc. A constructive proposal, untested; perturbations not done.

Item	Number	Counts against it / status
Stiff early component	$\Omega_{\rm stiff,0} \lesssim$ 4e-25 ($\Delta N_{\rm eff} = 0.5$), identical on both footings	A bound, not a prediction; the same charge sources dust and stiff, a fine-tuning cost.
BBN dark-field check	max $\Delta N_{\rm eff}$ = 2.0e-4 against a $\leq$ 1e-3 check; ΔY_P = -1.56e-10; Δln(D/H) = -1.60e-09; $N_{\rm eff}$ unchanged in the Y-gate branch	PASSES. The Li-7 4.0σ ($\times$2.95) overprediction is inherited from standard BBN. The control K2 failed (uncalibrated Y_P 7.2% below PRIMAT; $\Delta N_{\rm eff}$ response agrees to 9% after calibration).

Relics, sterile species and Koide-type relations: all closed or negative

Item	Number	Status
Relic neutrinos as the cold sector	phase-space m $\leq$ 93 eV against free-streaming m $\geq$ 148 eV; Tremaine–Gunn m $>$ 65 eV against $\Sigma < 0.1$ eV (650$\times$ short); mass budget 206$\times$ short	closed (a kill, not a prediction)
Hot relic and sterile routes	one fermion: clusters need m $\geq$ 14.68 eV, dwarfs $\geq$ 93.26 eV; thermal 1 dof gives $\Omega_{\rm dm}$ at 11.3 eV but $\Delta N_{\rm eff} \approx 1$ (about 6σ against 2.99 $\pm$ 0.17); the surviving corner is a 5–11 eV non-thermal window or a 100–200 eV relic at $T_x/T_\nu \approx 0.37$ that misses the CMB floor by 0.18 dex	keV sterile retracted as a gate bug; 11 eV sterile killed; the non-thermal window has no mechanism
No-neutrino reading (CFG254)	$N_{\rm eff} = 0$ against 2.99 $\pm$ 0.17 is +17.6σ	contradicted
Neutrinos as cold energy (CFG507)	$\Sigma = 0.0614$ eV is hot; a seesaw relic m* = 2.34 TeV gives Ωh^2 from 2.5e-4 to 112 over 144 forms, and 34.5% of 200 scrambled draws also hit	a restatement; it needs extra early energy $\leq 0.3\ \Delta N_{\rm eff}$ at BBN
Neutrino Koide	$Q_\nu = 0.585254$ at $m_1 \to 0$ against $\sqrt{(2/Z)} = 0.587788$ (-0.431%); 0.382 at m_1 = 10 meV; a free rational 7/12 fits better (0.33%). Charged leptons give 0.6667.	"glued, not forced"; no parameter-free neutrino Koide relation exists. The "shared 2/3" claim is 7.3σ off and discarded.
Scale-gap mechanisms	Addazi–Meluccio k = 0.255; Jia k = 5.01; neutrino-loop dark energy k in [46, 100]; McCulloch k $\approx$ 200; the dS / Planck see-saw gives 5e-3 meV, off by 1e5$\times$; thermal dS floor $kT_{\rm dS}$ = 1.9e-31 meV, about $10^{31}\times$ too small	none forces an absolute m_ν; all leave k free. The ${\rho_\Lambda}^{1/4}$ match is FDR-trivial (64% of the O(1) pool lands in 5–120 meV)
Radiation drag, inverted-BH ν_R, PMNS pre-registration	drag about 1e-19 a_0, relic push under 1e-6 a_0; SO(10) 16th state is an input; candidates in the PMNS windows were 26,142 / 15,212 / 20,799, chance survivors 2,614 / 5,071 / 4,160, all landing at 0.00σ	rejected; withdrawn as a derivation; withdrawn as unfalsifiable
Not found anywhere in the repo	no sterile-neutrino prediction, no mixing-angle derivation, no supernova or neutrino-clustering test, no tau-mass or Dirac mass-ratio relation, no neutrino isocurvature	—

Standing cautions and closed doors

- Σ **is hot, not cold energy.** A 0.0613 eV neutrino sum is hot dark matter and cannot be the cold energy (CFG507).
- **A coherent chain is not evidence.** The lepton chain (TM1 + Dirac + $m_1 = {\rho_\Lambda}^{1/4}$) is Lean-certified for its implications only (LEPTON_CHAIN 1–3, MODULAR_S4_FIXED; 10 theorems, zero sorry). Lean does not certify TM1, Dirac neutrinos, the coincidence, or κ.
- **Closed doors.** Residual symmetries (FS01–03), textures (FS06/13c; the one survivor A2 gives $\Sigma \approx 0.065$ eV, now above DESI and forbidden by the Dirac argument), modular A4 (FS07), modular S4 fixed points (FS17: TM1 is a 24/48 coin flip), swampland $\times$ evolving dark energy (FS18, not crispy), Koide routes. Do not re-run.
- **Scale gap, not mechanism.** The neutrino is the one SM sector where the scale gap closes (about 1 order, against 10^8 for the electron), but the closing is generic and mechanism-free. Radiation drag as the source of a_0 is about 1e-19 a_0: dead.

- **Older neutrino numbers.** The Z^2-era values (m_1 about 1.5 meV, Σ 58–60 meV, δ_{CP} 195°) conflict with this chain; see the K4 rows. They are not pooled.
- **The $a_0(z)$ coupling.** The tower $m_\nu(z)$ and growth-step rows ride the same dark-energy gate (G5) as the $a_0(z)$ test; the June deposit behind them (DOI 10.5281/zenodo.20977421) used the retired modified-inertia framing, though the evolving-ρ_{DE} logic does not need it.

5 K4: Kepler slop

K4: Kepler slop (speculative and numerology-era predictions)

Dated and testable, so they could still land, and several are further out. None has a derivation, and the record has retracted or downgraded most of them (the 'Z^2' era, $Z^2 = 32\pi/3$). They are on the chart because they could be checked, not because the framework stands behind them. Never cite them as framework results. Where a K4 value conflicts with a K1–K3 row, the K1–K3 row stands and the two are never pooled.

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
LiteBIRD / CMB-S4 *date not on record*	**Primordial tensor-to-scalar ratio r** $r = 1/(2Z^2)$; current bound $r < 0.032$–0.036. Includes the primordial gravitational-wave cross-polarization $h_\times = 0$	$r = 0.0149$ (0.014921).	**Rivals.** Not stated in the file. The current bound is $r < 0.032$–0.036; CMB-S4 and LiteBIRD target δr. **Counts against.** r below 0.010 or above 0.020 rules it out. The file concedes it is forward-testable and sits just under the current bound. **Note.** Status: the record calls it numerology. The Z_2-projection argument is wrong (both modes are Z_2-even) and the author's own file retracts it; the $h_\times = 0$ prediction dies with the same error. *real_research/reviews/tensor_scalar_r_audit.py; REAL_SCIENCE_RECOVERED.md; AI_SLOP_SALVAGE_REVIEW*	K4
LiteBIRD *date not on record*	**Cosmic birefringence angle** $\beta = 0°$	$\beta = 0°$.	**Rivals.** A new analysis suggests the rotation may exceed the widely quoted EB value of about 0.3°; vacuum-interface mechanisms predict nonzero β. **Counts against.** A confirmed nonzero β. The data already sit at 4–6σ tension (4.9σ in one review). **Note.** Status: derived, but already in tension, and it is just the Standard Model default, so it does not distinguish the framework. *OPUS_INDEPENDENT_REVIEW; z2-briefing-2026-06-04*	K4
JUNO *~2027–28*	**Neutrino splitting ratio** $\Delta m^2_{31}/\Delta m^2_{21} = Z^2$	33.51 ($Z^2 = 32\pi/3 = 33.5103$).	**Rivals.** Measured 32.6 ± 1.0 today, about 1σ (2.8% off). **Counts against.** A ratio measured away from 33.5 at JUNO precision. **Note.** Status: the earlier review called it the 'most interesting' match because it was not obviously cherry-picked. It remains numerology: no mechanism fixes the ratio. *OPUS_INDEPENDENT_REVIEW §2.4*	K4
NOvA / T2K, DUNE *ongoing to 2030s*	**CP phase** $\delta_{CP} = 13\pi/12$ from the gauge-count formula 180° (G+1)/G, G = 12	195°.	**Rivals.** NuFIT 6.0 IC19 gives 177 +19/-20; IC24 gives 212 +26/-41. 195° sits inside both. The K3 lepton chain predicts 262° or 285° instead. **Counts against.** δ below 180° or above 210° at 3σ. **Note.** Conflicts with the TM1 chain (262° / 285°). Do not pool. The site's own route notice treats this card as contradicted. *ai_slop/website predictions #5; OPUS review*	K4
SPT-3G / CMB-S4, DESI *2026–2030*	**Neutrino mass sum, older formula** Σm_ν from $m_e\,\alpha^3$ / 4 and a Z-spaced hierarchy	$\Sigma \approx 58.2$ meV from the file's formula (m_3 49.6, m_2 8.57, $m_1 = 0$; the site card prints 58 meV, the ledger says 59–60). An earlier table has $m_1 \approx 1.5$ meV.	**Rivals.** The K3 chain gives 0.0613 eV with $m_1 = 2.24$ meV; minimal normal ordering gives 0.0587 eV. **Counts against.** Σ below 50 meV or above 70 meV. **Note.** Conflicts with the $m_1 = \rho_\Lambda{}^{1/4}$ row; they cannot both be right. A Σ near 58 meV is also the minimal normal-ordering floor, so it cannot be distinguished. *ai_slop/website predictions #8; OPUS review*	K4

When	Test	Framework predicts	Rivals; what counts against it; status	Grade
HL-LHC *2026–27 (site)*	**Higgs self-coupling** $\lambda = (G+1)/(G-2)^2 = 13/100$	$\lambda = 0.13$.	**Rivals.** The Standard Model value, so it does not separate anything. **Counts against.** λ below 0.11 or above 0.15. **Note.** Equals the SM value, so a pass is not distinctive. *ai_slop/website predictions #7*	K4
LSST time delays *2026–27 (site)*	**Hubble constant** $H_0 = 71.5$ km/s/Mpc, between Planck and SH0ES	71.5 km/s/Mpc.	**Rivals.** Between the Planck and SH0ES values. **Counts against.** H_0 below 69 or above 74. **Note.** No derivation behind the value. *ai_slop/website predictions #9*	K4
JWST NIRSpec *2026–27 (site)*	**BTFR offset at z = 2, the H(z) form** $\Delta\log M_{bar} = -\log_{10} E(z)$	-0.47 dex at z = 2.	**Rivals.** This is the rival $a_0 \propto H(z)$ law. The record's distinctive flat law predicts 0.00 dex. **Counts against.** Offset above -0.3 dex at z = 2. **Note.** Contradicts the K1–K3 flat-a_0 rows, which say a_0 does not evolve below $z \approx 5$. Kept only to mark that it is the rival. *ai_slop/website predictions #1*	K4
SPT-3G + ACT *2027 (site)*	**High-z cluster counts** Enhancement set by E(z)	Enhanced by about $E(0.8) \approx 1.6$ (the quick table says 2×) at $z > 0.8$.	**Rivals.** ΛCDM counts. **Counts against.** Cluster counts match ΛCDM exactly. **Note.** Tied to the $a_0 \propto H(z)$ picture. No derivation. *ai_slop/website predictions #10*	K4
ACT + Planck void catalog *2026 (site)*	**Void kSZ outflow velocity** Vertex outflow	265 km/s.	**Rivals.** Not stated. **Counts against.** Below 200 or above 350 km/s. **Note.** No derivation. *ai_slop/website predictions #13*	K4
Euclid DR1 *2026 (site)*	**Lensing mass equals baryonic mass** No dark matter	$M_{lens} = M_{bar}$.	**Rivals.** NFW halos. **Counts against.** Clear NFW profiles. **Note.** Contradicts the standing record: the cold energy's mass is required and no cold energy breaks X-COP, the Bullet cluster and Local Group timing. *ai_slop/website predictions #3*	K4
DESI Year 3 + BAO *late 2026 (site)*	**Redshift-dependent S8 tension** Qualitative	The S8 tension evolves with redshift.	**Rivals.** A constant tension. **Counts against.** The tension resolves with no z-evolution. **Note.** The standing record makes no S8 prediction either way. *ai_slop/website predictions #2*	K4
SDSS / DESI quasars *2026–27 (site)*	**Topological duplicate images (T^3/Z_2, 20.6 Gpc)** Ghost quasars at 20–60° separation; a claimed DESI 4PCF 'confirmation' r = 0.9986	Duplicates at 20–60° in $z > 3$ quasars.	**Rivals.** No duplicates. **Counts against.** None found. **Note.** Excluded by the record: the η-invariant is exactly 0, CMB matched circles exclude the topology, and the ghost candidates are SDSS aliasing artifacts. The claimed 4PCF confirmation is not supported; do not cite. *REAL_SCIENCE_RECOVERED.md; reviews/ghost_quasar_sdss_live_check.py*	K4
precision constants *already measured*	**Standard Model constant 'hits'** $\alpha^{-1} = 4Z^2 + 3$, $\sin^2\theta_W = 3/13$, $\Omega_\Lambda = 3Z/(8+3Z) = 13/19$ and about 50 more	$\alpha^{-1} = 137.0413$; $\sin^2\theta_W = 0.2308$; $\Omega_\Lambda = 0.6846$ (Planck 0.6847 ± 0.0073).	**Rivals.** The measured values. **Counts against.** None: they are retrodictions. **Note.** FDR audit: an arbitrary target is hit this closely 19.9% of the time. Several constants carry two incompatible formulas. Numerology. *REAL_SCIENCE_RECOVERED.md; deepseek E10*	K4

6 Already confronted

What the record says so far. "Not diagnostic" means the test could not separate the readings, which is different from a pass.

Passes and agreements

Test	Outcome	Result	Source
SPARC rotation curves	Kernel $\nu_{\rm mono}$ rms 0.100 / 0.099 dex; band [7.8e-11, 2.0e-10] contains both footings. Non-diagnostic of a_0 itself.	pass	GATES 1.01; concordance P1
BTFR zero-point	Both footings inside [8.7e-11, 1.55e-10]; canonical about 1.8σ below the median.	pass	concordance P3
KiDS lensing RAR	χ^2/dof 2.03 / 0.94; canonical inside only via a hot-CGM file (marginal).	pass	concordance P2
Milky Way vertical force	Round cold energy beats the QUMOND phantom disc in 32/32 cells; MUTATE restores the rejection.	pass	CFG514/516
Early-type lensing levels, clusters, MW satellites	Pass with the census edge: lensing -0.05 / -0.12; clusters blind and declared; satellites 2/39 unbound.	pass	CFG515
Baryonic Faber–Jackson, groups	Consistent supply closes the gap: +0.00 to +0.06 dex.	pass	CFG540/543
$c_T = c$, $\gamma_{PPN} = 1$, solar-system nulls	GW170817 exact; lensing = dynamics residual 0.601σ; planetary nulls pass by 3.8–6.8 orders. The covariant door fails Cassini Q2 by 13–17σ (retired door).	pass	concordance N1–N4
CMB, Lyα forest, SN host-mass step	CMB $\Delta\chi^2$ 1.343 / 4998 multipoles; Lyα 0.4–0.9σ; 6.9σ step reproduced (decisive tests 18% power).	pass	ledger B
Local a_0 and RC100 flat law	47 MeerKAT discs match SPARC; RC100 flat a_0 disfavours the ΛCDM-native rise at 3.9σ (a restatement of the falling dark-matter fraction).	pass	CFG301; SLH 3rd pass
No bar-strength dependence, bulk-flow null	-0.026 ± 0.039 dex; $\beta = 0.447$ against 0.440.	pass	SLH

Fails and tensions

Test	Outcome	Result	Source
Structure growth, gravitating field	Fails at 512^3: 0.150–0.193 against the 0.10 cut, σ_8 +3–5%. The particle-density pass stands (PAPER45 v2.2). The excess is intrinsic and persists at about half size with the measured supply (0.66 → 0.29 can).	FAILS	CFG555/556/595
Milky Way outer slope	Z -3.6 / -3.9 (can / alt): shape excluded as frozen, resting on the distance scale. DR4 decides.	FAILS	CFG532b
SLUGGS central ellipticals	+0.07 / +0.06 dex after the joint best case (class 3.5σ / 2.8σ, statistical only).	FAILS	CFG528
Milky Way ultra-faints	+0.325 / +0.304 dex, 3.8 / 3.5σ against candidate B.	FAILS	GATES 1.09
Coma ultra-diffuse galaxies	+1.159 dex (factor 14), 4.9σ can / 4.7σ alt. Never cite 19.4σ.	FAILS	standing rules
M31 satellites, Local Group timing	M31 LVD -2.67σ. LG timing $z_{\rm full}$ +2.2 can (strict radial +4.4), a data-limited tension shared with ΛCDM.	FAILS	CFG286; CFG522/548
Chae EFE fits (the cost of R7)	4.1 / 4.3σ from zero external field.	FAILS	GATES 1.15
a_0 rising with z: MUSE-DARK III and eRASS1	MUSE $a_0(z\approx1) = 2.38$e-10, a rise (15.1σ raw, 1.9–3.0σ after drift); the standing notes call it a live threat. eRASS1 η(z) +0.187 ± 0.013 dex per unit z (selection-prone) against zero for constant a_0.	FAILS	standing rules; Ciocan+26
No cold energy	Breaks X-COP (13–15σ), the Bullet cluster (Z_B +7.9) and LG timing; SPARC rms 0.100 / 0.101 dex. Mass is required.	FAILS	CFG624
Reionization optical depth	τ_e 0.064–0.070 against Planck 0.054 ± 0.007 (+1.4–2.2σ, adverse).	FAILS	NF P19
Retired-arm failures	Single-metric MI lensing under-lenses (F(1) 0.563 vs $1/\nu$ 0.707); bulk-flow boost overshoots about 10×; the $a_0/2$ planetary tail is excluded ×1018 to ×40,000; the a_0/c drift ×246 (MESSENGER); the disformal photon metric by 7.3 orders. These closed the modified-inertia arm.	FAILS	prep_2026 SALVAGE_SYNTHESIS, DOOR_SCOREBOARD

Not diagnostic

Test	Outcome	Result	Source
Cosmic shear, framework-native	$\Delta\chi^2$ +23.85 / +4.20 (can), +30.04 / +5.56 (alt) on 20 KiDS / 38 DES points. Not diagnostic under the frozen rules; leans against. The S_8-free fit needs 0.746 / 0.726. CFG590 and CFG591 are context only (ΛCDM-calibrated machinery).	not diagnostic	CFG592
KiDS inner shortfall	ϵ +0.34 / +0.23 at 52–98 kpc; footing-dependent (alt closed by a +0.10 dex stellar mass). The edge position does not move it (CFG600, void); the leaked-satellite normalisation does (CFG601).	not diagnostic	CFG529/531/600/601
Early-type age signal	Raw KiDS, equal-mass old vs young early types: K9 A = +0.166 (×1.46), Z +2.33; stellar-mass bias not excluded. Colour clock undecided; early-only (Z +3.44 matched).	not diagnostic	CFG585–589, 700
High-z a_0 ($z \approx 1.5$–5)	KURVS lean to the rival, flat under Burkert, CRISTAL both-consistent, ALPAKA / Amvrosiadis / Roman-Oliveira lower bounds, BX442 pressure-dependent (a_0 ×2.4), JWST-baryon discs y = 1–48. None is a detection.	not diagnostic	CFG140–229, 270–285, 437, 602
High-z BTFR, SNe a_0	Jeanneau Δb +0.140 ± 0.272 (0.51σ can, 1.41σ alt); footing fork a wash (0.99 / 0.95σ); SNe ρ_{DE} against galaxy a_0 ratio 1.64, +1.5σ.	not diagnostic	jeanneau_refit; sne_lambda
Cosmic-time and dark-energy triangulation	$f_{DM}(<R_e)$ decline -0.086 predicted vs -0.080 ± 0.035; H(z) tension Z 2.2–3.2, route-dependent.	not diagnostic	CFG547
Merger-vs-galaxy settling	$\alpha = \{0\}$ only on both footings; α = 0.1, 0.3, ∞ excluded (Z_B +5.9 to +8.1). CFG597, 599 and 622 void.	not diagnostic	CFG621/623
UFD infall order, clusters vs MW cold column, bar friction	Wrong sign none reaches 2σ; suppression below 20% at 95%; partial ρ -0.091 (p 0.35).	not diagnostic	CFG463/517/560
MW outer HI flaring, DiskMass σ_z, SAGA, void lensing	No qualifying measurement or non-discriminating.	not diagnostic	CFG561–578
Wide-binary DR3 dry run	$\hat{\gamma}$ 1.205 ± 0.035 (can) / 1.2025 ± 0.0337 (alt), contaminated: evidence for nothing.	not diagnostic	PREREGISTRATION_DR4
Bullet-type cluster mass	Lensing mass traces galaxies (right sign); the kernel supplies 2.6–3.3× against 8.3–9.8× needed, a 3.2× gap.	not diagnostic	NF P5

Added: further results

Test	Outcome	Result	Source
Local Group and group zero-velocity radii	R_0 = 1.28–1.62 Mpc against 0.96 ± 0.03 (+0.13 to +0.23 dex); isolated MOND 1.93 / 2.02. The overshoot is shared by LG, M81 and IC 342 (+0.20 dex; 14-group stack +0.12 to +0.14). ΛCDM shares it.	FAILS	XR4_data_gates; ANSWER_AS_IT_STANDS
Merger infall (Bullet, El Gordo)	The split-force carrier fails load-bearing checks A3, B3, B6: El Gordo needs v ≈ 2.09–2.29 against ΛCDM's 1.44 (Jubilee max 1.69).	FAILS	merger_infall_2026
Acceleration-trigger carrier	Keeps the flat a_0 zero point within 0.074 dex but fails cosmic shear at every KiDS-safe cap $\geq$ 1.75 Mpc; all constants hand-set.	FAILS	acceleration_trigger_2026
Binary galaxies	$\sigma_{los} = 0.60679\ (G\ m\ a_0)^{1/4}$; on 1900 pairs A = 1.802 ± 0.041 (19.6σ), 2.311 with EFE; deficit grows with mass at 3.9σ.	FAILS	FINDINGS L617-659
Pressure-supported systems	The M–σ coefficient is about 2.5× the theorem's 4/81; isolated dwarfs +0.06 to +0.18 dex, MW satellites +0.23, globulars -0.3.	FAILS	hunt_2026 f14/f15
No-dark-matter CMB face	Peak 3/peak 2 = 1.192 against ΛCDM 0.992, -20.3%, about 20σ.	FAILS	FINDINGS L295
Cassini quadrupole, QUMOND exponential kernel	Q_2 = 1.9562e-26 s^{-2} = 3.76× the Park ceiling (can); 2.5802e-26 = 4.96× (alt).	FAILS	EXACT_EXPONENTIAL ledger
EFE in dwarfs and cluster-infall BTFR	Dwarfs 3.87 / 3.90σ (4.2–4.5σ with hot gas), cluster slope 2.4–2.9σ, zero point 2.8–3.1σ, combined about 4.5–6σ against zero EFE (the cost of R7).	FAILS	XR4_data_gates
MW LMC reflex	3e9 $M_\odot$ of LMC baryons cannot produce the 32 km/s reflex (off ×4.9); the MW azimuthal asymmetry is under 1.43% (underpowered).	FAILS	hunt_2026 h81/h82
Cold-budget lane (T15, AI branch)	The settled kernel exceeds the observed deficit 2.2× (groups) and 3.5× (clusters); no single universal coupling survives the cluster budget (3.8×); coupling must fall 1.6–3.8× from galaxy to cluster.	FAILS	OMX t15, t16
Gas-rich dwarf BTFR scatter	A semi-analytic ΛCDM-emergent a_0 predicts 2.2–2.4× the observed scatter.	FAILS	THE_THEORY_AS_IT_STANDS:325

Test	Outcome	Result	Source
SPARC a_0 by population	Full sample optimum 1.37e-10 (9.36e-11 disfavoured at about 3.3σ, $P < 0.002$); gas-dominated cut (N = 37) 9.8e-11. PAPER43: 8.3 ± 1.1 e-11 (1.2e-10 disfavoured 2.7σ). PAPER40: 1.31e-10 (recipe ±0.128 dex). The footing is not decided.	not diagnostic	WHICH_A0_POPULATION_SPLIT; PAPER40/43
Big Wheel z = 3.25	a_0 rising as cH(z) is disfavoured at 1.9σ (framework footing) to 2.3σ (empirical); $a_0$eff 1.0–1.6e-10 against a rising 4.63 / 5.94e-10. A weak, negative-only result.	pass	BIG_WHEEL_A0Z_2026.md
MUSE-DARK III against constant a_0	Alcock fit exponent 0.78 ± 0.15; frozen a_0 excluded 5.3σ (6.6σ with anchor). The record reads it as ΛCDM-native (NFW a_s ×2.57 at z = 3), so it is contested and degenerate.	FAILS	A0Z_ALCOCK_COMPARISON
tSZ outer slope (blind map test)	X-COP GNFW gives -2.48 ± 0.33 against a registered window (-1.7, -0.9); the window was rewritten before any map was read; data predate registration. Not passed, not killed.	not diagnostic	tsz_score_2026
Rotation-vs-pressure matched pair	+0.215 dex (alt +0.205), 1.73σ: fails the 3σ bar; a weak hint.	not diagnostic	hunt_2026 F08_F10
Outer residual against baryonic mass	Slope -0.072 in WALLABY (85 discs): contradicted, marginal.	not diagnostic	CFG584
EFE break radius	Spearman -0.811 (MOND, by construction) against -0.028 for NFW; 34 outer points of 9–10 galaxies fit worse (0.304 vs 0.135 dex). SPARC cannot decide.	not diagnostic	FINDINGS L245
Early-type RAR phantom ellipticity	Projected q = 0.971 for a disc; early types f_h 0.151–0.214, ΛCDM 0.314, measured 0.217 ± 0.160 (blue) / 0.303 ± 0.080 (red). Consistent with everything.	not diagnostic	hunt_2026 h62

7 Published deposits

What is on the public record with a DOI or a committed hash, and where each stands.

Deposit	What it registered	Status	Source
PAPER35: wide binaries exactly Newtonian	DOI 10.5281/zenodo.23142564, recorded 2026-10-04. $\gamma_v = 1.000$ on the pre-registered DR4 estimator; the same law with an external field gives 1.16–1.23 (Arm A's floor).	open	qwen papers_2026/PAPER35
PAPER7: flat $a_0(z)$, the sharp null	Pre-registration DOI 10.5281/zenodo.22563139 (2026-09-06); v3 22833314. BTFR zero point 0.00 dex at z ≈ 2.5 against ΛCDM +0.33.	open	PAPER7
PAPER14 v2: observing case at z ≈ 2.5	DOI 10.5281/zenodo.22961490 (2026-09-25). Gate window z ≥ 2, lens gate 0.05 dex; needs 3 galaxies at ±0.10 or 4 at ±0.20. v1 superseded (a ledger row contradicted its sources).	open	PAPER14
PAPER38 v1.3: $a_0(z)$ calibration wall	DOI 10.5281/zenodo.23101137 (v1.1 23073072, v1.2 23085582). Flat $a_0(z)$ as the distinctive high-z prediction; v1.3 corrects the DR4 paragraph to the P2 kernel, 1.063–1.102 (can). v1.1–v1.2 DR4 numbers superseded.	superseded	PAPER38
Rubin w(z) gate	DOI 10.5281/zenodo.21478568, frozen 2026-07-21. $R = a_0(3)/a_0(0)$ about 0.60–0.75; verdicts OPEN / DISSOLVED / STRAINED.	open	rubin_prereg
Oort-cloud anisotropy	DOI 10.5281/zenodo.21966646. Spike modulation 1.069, injection 1.169.	open	ledger A-5
Gaia DR4 pre-registration, amendments 1 to 21	DOI 10.5281/zenodo.21702746 (pre-Amdt 11). Amendments 11–21 filed pre-data; targets from Amdt 1–9 are superseded.	open	gaia_dr4_prep
PAPER45 v2.2: supply-edge note	DOI 10.5281/zenodo.23240863 (v2.1 23240853; concept 23219294). v2.2 corrects the growth claim: all four 512^3 runs fail the 10% cut (P ratio 1.0405 / 1.0459 / 1.0429 / 1.0333); 256^3 passes.	superseded	PAPER45
PAPER42: Λ fixes the turn-off	DOI 10.5281/zenodo.23171066. $y_t = 94$ (74–120), near 1e-8 m/s^2; structurally out of reach (about 0.002 dex). v1.2 corrected the 2σ band to 1σ.	note	PAPER42
PAPER43 and PAPER40: a_0 measurements	DOI 10.5281/zenodo.23175719 (8.3 ± 1.1 e-11) and 23142559 (1.31e-10, recipe ±0.128 dex). Measurements, not forecasts; footings never pooled.	done	PAPER40/43
PAPER44: EFE-blind MOND	Draft 2026-10-06, v1.1 10-08; Zenodo script prepared, not run. γ_v 1.20 at 10 kAU, 1.51 at 20 kAU. CFG447 says Gaia DR3 excludes this reading.	superseded	PAPER44
Atomos null (v2, v3)	DOI 10.5281/zenodo.21654272 (v2) and 23246734 (v3, correction release). 174,890,804 raw and 42,534,139 distinct combinations, 0 certified. Twelve v2 numeric claims were wrong; the null stands.	done	project_atomos
Data-gate pre-registration	DOI 10.5281/zenodo.20935948 (2026-06-26). Gate 1 (Euclid WL-RAR slope 0.5) survives as a MOND-class test; Gate 2 superseded by the DR4 amendments; Gate 3 (sˆTX) void.	superseded	DATA_GATE_PREREGISTRAT

Deposit	What it registered	Status	Source
Modified-inertia era deposits (June–July 2026)	s^TX note 20978308; dwarf orbital-history σ 20949773 (v3 20969585); sign-flip 21179352; σ-spread 21421896; SHLEM 21458606; TDG history 21461435; cross-scale $a_0(z)$ 21440408. All superseded with the MI arm; the retractions file says lensing excludes pure MI at 21σ.	superseded	RETRACTIONS.md
Sealed hashes in the repo	CFG553 (Milky Way, f8bea44f…), CFG571 (ALMA, be27457d…), CFG549 (dark energy, f110566e…). Never edited after the seal.	open	PREDICTIONS_HASH.txt files

8 Claims not to cite

Withdrawn, superseded or void claims that still sit in older files of the programme.

- **s^TX amplitude and bands** 8.68e-10 collapsed to 1.258e-15 under the $\alpha = 2$ kernel (Amdt 5). The '9.6×' margin is also void.
- **Wide-binary targets 1.09, 1.0246, 1.0310, 1.1582, 1.2139 / 1.2592** Superseded by Amdt 10 to 14. The gated 1.0004–1.0006 'Newton confirms the framework' was withdrawn (Amdt 6).
- **$a_0(3) = 0.74$, +6% bump at z ≈ 0.4, CPL a_0** Retired dressing. The derived law is flat below 1% for $z \leq 5$. The CPL numbers survive only as one dark-energy-tracking branch.
- **'$\kappa = \frac{1}{2}$ derived'** κ is fitted. The deepseek PD01–22 and glm53 'equilibrium' claims, and GF2's SIS ratio, are not the law's κ.
- **'Theory closed' or 'every door closed'** Say every local route is closed and one nonlocal door remains.
- **Ly-α '6–8σ' exclusion** Corrected to 0.4–0.9σ.
- **Coma UDG '19.4σ'** Use 4.9σ (can) / 4.7σ (alt) / 2.7σ (first infall).
- **5.09 keV line, the 2.55 keV line, $\lambda_{fs} = 0.558$ Mpc, 'cosmic-noon freeze'** Self-anchored; 6.3σ below the forest floor. Do not list as predictions.
- **KiDS 'tolerates 6–14% of a CDM-like halo'** A covariance bug. Void.
- **MUSE reconciliation; v9 dark sector; condensate or cluster phase-pinning dark sector** Dead or withdrawn.
- **Atomos v2 claims** The no-valid-holdout and 'Gate B forced' claims are withdrawn; v3 (DOI 23246734) is current.
- **'Z as the dark-to-baryon ratio'; MIGHTEE '5σ' a_0 rise; 'DR4 a_0 20–21% worse than SPARC'** Wrong or flips with M/L (DR4 is 12.6% better).
- **Priority over Alcock 2609.38796** Never claim Zenodo priority; his letter was received 07-10.
- **Council sessions and the 'light-end soliton' for UFDs** Fictional AI transcripts in ai_slop/. Only the session RESULTS files are real.
- **Older ledger wording** The 09-02 and 09-09 ledgers and the June-era PREDICTIONS_100 predate the retractions. Re-check every number before citing.
- **'Z^2 confirmed' claims on the public site** The route notice and several cards still say DESI 4PCF 'confirms' the T^3/Z_2 topology (r = 0.9986) and list the Z^2 constants as live. The η-invariant is exactly 0, matched circles exclude the topology, and the record retracts the constants as numerology. See the K4 rows.
- **BH-star and little-red-dot 'coincidence'** Audited 09-22: not a framework result. The pulsation ceiling is off by about 1.9 dex (6.56e7 against a registered 1e5–1e6 M⊙).
- **DeepSeek deep-end SPARC rows** Built from kinematic M/L fits. With 3.6 μm ratios the '6% agreement' disappears (a_0* ratio 0.44–0.52 against 0.94); do not quote those rows as passes.
- **Public site /predictions page** The quarantined ai_slop page still lists $Z^2 = 32\pi/3$ numerology as live cards ($\delta_{CP} = 195°$, $\Sigma m_\nu \approx 59$–60 meV, $\lambda_{Higgs} = 13/100$, $H_0 = 71.5$). The site's own route notice contradicts it.
- **Public site wording on $a_0(z)$** The layout says a_0 is flat below z ≈ 20 and any robust evolution at $z \lesssim 5$ falsifies it. The 10-10 kill sheet (G5) is the dark-energy-tracking version (+6% near z 0.4, about 0.74 by z = 3, flat only if w = -1). Reconcile before sharing.
- **Older DR4 value text** '1.16–1.18 merge' belonged to Arm A. 'Newton confirms the framework' and the 4.74–7.10σ / N ≈ 1157 line are stale.
- **PREDICTIONS_100** 104 items: 33 'distinctive' and 71 shared-MOND. The live non-shared items restate the $a_0(z)$ ladder (1.009 / 0.862 / 0.737 / 0.641 / 0.566 / 0.506 at z = 1–6, CPL-era); three are the retired bump.

9 What the framework does not predict

Statements the framework does not make, listed so they are never claimed as tests.

- **S8** No prediction either way. Linear order is inherited; the nonlinear sign is adverse and unpriced.
- **Early-galaxy counts at z > 10** Halo abundance is ΛCDM's, since the law switches off before turnaround. The JWST UV-luminosity excess is neither help nor harm.
- **21-cm and cosmic dawn** No linear signature, though the transition falls in the EDGES window.
- **Collapse-timing speed-up (1.1–2× at z = 6–25)** Real but observationally inaccessible; it needs rotators below Σ† at z > 5.
- **Separating B from Newton or ΛCDM in DR4** All three predict $\gamma = 1.000$. A Newtonian result is survival, not confirmation.
- **a_0 from Gaia DR4** DR4 does not measure $a_0 = 9.36\text{e-}11$.
- **Killing ΛCDM** No committed forecast separates the framework from ΛCDM above 1.3σ.
- **The Standard Model mass sector** Walled. No forced structure; textures A2 withdrawn.
- **Clusters alone** The framework does not close clusters by itself; η(R500) 1.72–2.08 is a standing liability.
- **Self-assessment from the data census** 'Viable, but essentially no distinctive, MG-impossible in-hand evidence.' DESI DR3 w = -1 is the pre-registered kill for the $a_0(z)$ variation.

10 Limits and use

This ledger is a record of what the programme has committed to, not an argument that the framework is right. Several rows are conditional on assumptions that are not derived ($\kappa = \frac{1}{2}$, the supply of cold energy, the identification of the lightest neutrino mass with the vacuum-energy scale, the non-supersymmetric AdS conjecture). The programme's own census describes the framework as viable but with essentially no distinctive, modified-gravity-impossible evidence in hand. A survival of a test is never a confirmation. The dated sealed predictions (the Milky Way vertical force and curve, the ALMA gas masses, the dark-energy evolution statement) carry SHA-256 hashes in the repository and are never edited after sealing; any change goes in a new dated file.

AI assistance. The sweeps and this compilation were produced with Anthropic's Claude; the numbers were read from committed files, and the build script regenerates every table from the chart data. Rows from other AI models' exploration folders are labelled as such.

Reproducibility. The repository at `https://github.com/carlzimmerman/zimmerman-formula` holds the public chart (`explainers/kepler_grade_prediction_chart.html`), this paper's builder (`PAPER46_build.py`) and its audit (`PAPER46_audit.py`).